Tribhuvan University

Faculty of Humanities and Social Sciences

EventAura

A PROJECT REPORT

Submitted to

Department of Computer Application

National Infotech College

In partial fulfillment of the requirements for Bachelors in Computer Application

Submitted by

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National Infotech College

SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by Jyoti Kushwaha and Sonali Patel entitled "**EventAura**" in partial fulfilment of the Requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

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SIGNATURE

Anish Ansari

SUPERVISOR

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National Infotech College

LETTER OF APPROVAL

This is to certify that this project prepared by Jyoti Kushwaha and Sonali Patel Sonali entitled "**EventAura**" in partial fulfilment of the requirements for the degree of Bachelors in computer application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

SIGNATURE of Supervisor	**SIGNATURE of HOD/ Coordinator**
Anish Ansari Software Engineer National Infotech College Birgunj, Nepal	Bijay Kushwaha M Phil National Infotech College Birgunj, Nepal
Internal Examiner	**External Examiner**

Acknowledgement

Firstly, we would like to thank National Infotech College for supporting and providing a proper platform and environment to work on this project. We would like to express our special thanks of gratitude to our supervisor Anish Ansari, who gave us the honor to do this wonderful project on the topic “EventAura”. We are also thankful for his continuous feedback and encouragement throughout the project.

We would also like to thank our respected sir Bijay Kushwaha for supporting and guiding us throughout this project. We are also overwhelmed with all the support and guidance that we got from our parents and friends. They have assisted us from time to time, providing us with feedback and encouragement. Lastly, we are thankful to all those who have always been encouraging and those who helped us overcome the challenges we faced.

Abstract

EventAura is a web application designed to simplify event management for organizations, venues, and users. Organizations can register, find venues, and organize events, while venues can list their locations and capacities. Users can register, select interests, receive event recommendations, search for events, and purchase tickets. Utilizing algorithms like social filtering for recommendations and Haversine for nearby event searches, EventAura aims to streamline the event management process, making it efficient and user-friendly.

Keywords: EventAura, Event Management System, Recommendation System, Location within a radius, Haversine Formula

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List of Abbreviations

CSS	Cascading Style Sheets
EC	Electronic Commerce
FRs	Functional Requirements
HTML	Hyper Text Markup Language
JS	JavaScript
NFRs	Non-Functional Requirements
PHP	Hypertext Preprocessor
UI	User Interface

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Chapter 1: Introduction

1.1 Introduction

EventAura is an innovative web-based event management platform designed to streamline and simplify the process of organizing and attending events. With a focus on enhancing user experience and efficiency, EventAura serves three main user groups: organizations, venues, and users. Organizations can leverage the platform to register, search for suitable venues, and create detailed event listings that include categories, dates, locations, estimated participants. This structured approach ensures that event organizers can efficiently manage and promote their events.

Venues, on the other hand, can register on EventAura to list their location and capacity, making it easier for event organizers to find and book appropriate spaces. This not only helps venues maximize their utilization but also provides organizers with a wide range of options to choose from. Users of EventAura can register to browse and search for events based on their interests, receive personalized event recommendations, and purchase tickets for paid events. The platform's recommendation system uses advanced algorithms like cosine similarity to suggest events based on user preferences and past interactions, while the Haversine formula enables location-based event searches.

For the development of this webapp, I am going to be using PHP and Javascript. It uses MySQL as database, PHP for backend development and Javascript for frontend development. HTML and CSS is going to be used for the website design. By integrating these advanced algorithms and a well-structured database, EventAura aims to revolutionize event management in Nepal, making it more accessible, efficient, and enjoyable for all users involved.

1.2 Problem Statement

- Lack of a centralized platform for organizations, venues, and users to manage and participate in events efficiently.
- Challenges in finding relevant events and venues within a specific location and interest area for users.

1.3 Objectives

- To develop a comprehensive and user-friendly web application that streamlines event management by providing personalized recommendations using cosine similarity and identifying nearby events or venues using the Haversine formula.
- Problem Addressed: Event organizers often struggle with low attendee engagement due to a lack of personalized recommendations and difficulty finding relevant suggestions and the Haversine formula to display nearby events, enhancing user satisfaction and participation.
- To implement secure user authentication and role-based access control, ensuring only authorized person can access or modify event information.
- Problem Addressed: Unauthorized access to event data can lead to security breaches. EventAura ensures only authorized personnel can manage event details, protecting data integrity.
- To enable users, organizations, and admin to effortlessly create through a collaborative platform that offers customizable features and a user-friendly interface.
- Problem Addressed: The process of event creation is often complicated and time-consuming, leading to frustration for organizers. By allowing users, organizations, and admin to easily create events with tailored features, EventAura simplifies the event management process, improving efficiency and fostering greater user engagement. similarity and provides nearby events or venue using haversine formula.

1.4 Scope and Limitation

Scope:

EventAura aims to serve as a versatile and accessible platform for event management across Nepal. It encompasses functionalities for event organizers to efficiently plan, promote, and manage events of various types and scales. Venue providers can list and showcase their spaces, while users benefit from personalized event recommendations and seamless ticket purchasing options. The scope includes integrating geolocation for venue searches and implementing algorithms for effective user engagement, ensuring a comprehensive solution for all stakeholders involved in event organization and participation in Nepal.

Limitation:

The main limitation of this project is initial focus limited to Nepal, potentially restricting global scalability.

This project is also going to be purely web-based.

1.5 Report Organization

The topic of our project is “EventAura”. It is divided into various chapter. In chapter 1 it includes various topics like introduction, objectives, problem statement and scope and limitation.

In chapter 2, it includes background, literature review and findings on similar topic.

In chapter 3, it contains technical terms as system analysis and design, requirement analysis, architecture and database designs etc.

In chapter 4, it contains implementation and testing of the project.

And chapter 5 concludes the project giving future recommendation.

And finally appendices contains the screenshots of the project.

Chapter 2: Study of Existing Systems and Literature Review

2.1 Study of Existing Systems:

There are several existing applications for event management. Some of them include:

Eventbrite: A popular platform for discovering and organizing events. It allows organizers to create, promote, and sell tickets for events of various sizes and types. [1]

Whova: Offers event management solutions including attendee management, agenda planning, and networking features. [2]

EventMobi: Provides event management and attendee engagement solutions through mobile apps and online platforms. [3]

Cvent: Known for its robust features, Cvent offers event management, marketing, and technology solutions. [4]

Whova: Provides a suite of tools for event registration, networking, and attendee engagement. [5]

Ticketmaster: A global leader in ticket sales for concerts, sports, and theater events. [6]

2.2 Literature Review

Regarding this project, I researched and review some of the related websites, articles, thesis, documentations, and applications. Though there are thrift stores across the globe, it hasn't been properly implemented online and in Nepal.

A. P. K. Pinjari and K. Nur on their research work, "Event Management System" is a software project designed to handle and manage various aspects of event planning. It includes modules for managing customer and employee information, event details, and services like location, transport, decoration, catering, and DJ. The system allows administrators to view, add, update, delete, and generate reports on customer and employee data. Administrators can also create e-cards for invitations and send them via email. Customers can check the status of their events on the event management website using their event and customer IDs. [7]

M. R. L. Perez, A. C. Lagman, and R. T. Adao on their research work, "The Event Management Solution" is a web application for EventAura Consultancy that helps users manage events, announcements, and website content. It includes modules for event management and registration, and it can generate easy-to-access reports from a centralized data repository. This helps reduce or eliminate inaccurate reports that were a problem in EventAura Consultancy's previous processes.[8]

A. Khan et al. on their research work, “Event Management System,” the event management system helps organize and manage all activities related to an event, including coordinating multiple service providers. It ensures that event organizers have the contact details of these providers for efficient communication. The software allows students to register for events online and access event details after logging in. The Head of Department (HOD) updates event information on the department portal, making it easy for students to get all the details. After the first round of an event, the system provides a list of qualified and non-qualified students. [9]

P. S. JosephNg et al. on their research work, “Design of a web-based platform: Event-venues booking and management system,” Organizing an event involves many tasks and managing finances, which are often done manually. The Event-Venues Booking and Management System aims to make this process more efficient by allowing people to find, book venues, and track expenses through a cloud-based web platform. Based on survey and interview data, the system's features are designed to meet users' needs. The findings show that this online platform helps users arrange and manage events more easily and quickly, reducing costs and time. Additionally, it fosters a positive relationship between users and suppliers. [10]

S. Caballe et al. on their research work, “Towards a platform-independent event management model for web collaboration,” Web collaboration enables virtual work teams to collaborate online using various tools like text chats, discussion forums, and document repositories. These tools support coordination, cooperation, and communication, with communication bEventAurag fundamental. Awareness is crucial in enhancing group motivation, interaction, and problem-solving, allowing for implicit coordination and spontaneous communication. This paper discusses knowledge management and awareness in online work teams and proposes a platform-independent event model for effective event notification, offering extensibility and reuse. [11]

C. L. T. Huyen on his research work, “Method and system for internet based event management at universities (Case study: Van lang University),” Organizing events in educational settings, like universities, is crucial but often suffers from poor information management, resulting in low student attendance. Traditional methods make it hard to check attendance, gather information, and collect feedback. This research examines the need for better event management at universities and proposes an Internet-based system. This new system, designed for use on mobile phones and computers, aims to reduce costs, streamline event management, and boost student engagement. Initially tested at Van Lang University, the system shows promise for broader application in creating smarter, more connected educational environments. [12]

C. Gigool, L. Gonsalves, and C. Correia on their research work, "Online event booking and management system," The event management system is designed to handle all activities related to organizing an event, a rapidly growing profession. Managing multiple service providers, such as sound systems, lighting, floral decoration, catering, platform construction, and venue booking, can be very challenging. This website helps streamline these tasks by allowing event managers to quickly mobilize their network of service providers, ensuring a successful event. The system aids in the efficient administration of these providers, making the entire process more manageable. [13]

D. A. Prasetya et al. on their research work, "Resolving the shortest path problem using the haversine algorithm." Route search requires an accurate algorithm to calculate distances between two points on the Earth's round surface. The Haversine algorithm is ideal for this as it calculates the distance based on longitude and latitude, providing precise measurements on a sphere. Using a simple formula and computer calculations, Haversine accurately determines distances between two locations. This research applies Haversine to find the closest location. Graphs are used to visually represent information, making structures easier to understand. The Haversine algorithm effectively calculates distances between Earth coordinates, helping to identify the shortest routes. [14]

E. Winarno, W. Hadikurniawati, and R. N. Rosso on their research work, "Location based service for presence system using haversine method," Location-Based Services (LBS) provide information about the position of a device or user, often via various media. The Haversine formula calculates the distance between two points and is applied in this research to a presence system based on location distance. This system uses mobile applications to record user presence and preserve data. LBS technology detects users within a specified area by identifying the closest point to a predetermined center point, which stores the coordinates and radius of the presence area. This research enhances the Haversine method for distance measurement and estimation, incorporating signal measurements with Wi-Fi and GPS to accurately identify user locations. [15]

C. N. Alam et al. on their research work, “Implementation of haversine formula for counting event visitor in the radius based on Android application,” Location-Based Services (LBS) use geographic data to track the presence of mobile devices. LBS can map information about areas, such as the presence of events, which attract more visitors as they gain popularity. The Haversine formula calculates the distance between two points and is used in this study to measure the distance between users and various events. By applying the Haversine formula, the system calculates whether users are within the radius of an event. If they are, the number of visitors to the event is updated. This method is implemented in an Android application, demonstrating that the Haversine formula can effectively calculate distances and determine event attendance. [16]

A. R. Lahitani, A. E. Permanasari, and N. A. Setiawan on their research work, “Cosine similarity to determine similarity measure: Study case in online essay assessment,” Technological advancements in education have simplified learning processes, file sharing, and assignment management. Automated Essay Scoring (AES) is a system developed to automatically score text documents, making grading more efficient and less subjective for lecturers. Implementing AES involves various factors, including language and scoring mechanisms. Tests on Indonesian text-based documents, after pre-processing for data extraction, result in a ranking of documents based on their similarity to the expert's document. [17]

P. Xia, L. Zhang, and F. Li on their research work “Learning similarity with cosine similarity ensemble,” Similarity is crucial in machine learning and pattern recognition, helping tasks like classification and clustering succeed when measured well. Cosine similarity, a simple and effective metric, is commonly used for this purpose. This paper introduces the Cosine Similarity Ensemble (CSE) method, which enhances similarity learning by using multiple cosine similarity learners. Each learner starts from a different initial point to define pattern vectors, not necessarily starting at the origin, ensuring diversity. The thresholds for these learners are adaptively set. The CSE method also incorporates a selective ensemble approach, and it outperforms other methods across various data sets. [18]

Z. Shen et al. on their research work, “Collaborative Filtering Algorithm Based on Contrastive Learning and Filtering Components,” Graph neural networks have emerged as a powerful tool for recommendation systems, particularly those involving social network graphs. By integrating social network data with user-item interactions, these systems can capture dynamic user preferences for more accurate recommendations. However, challenges like noise and data sparsity can hinder performance. To address this, a new model called FMPRec combines contrastive learning with filtering components inspired by signal processing. This approach reduces noise and enhances user-item embeddings, leading to better predictions with lower time complexity. Experiments show that FMPRec outperforms traditional methods in accuracy, validating its effectiveness. [19]

U. Shardanand and P. Maes on their research work, “Social information filtering: Algorithms for automating ‘word of mouth,” In the context of collaborative filtering and recommendation systems, previous research has highlighted the effectiveness of combining social network graphs with user-item interactions to improve recommendation accuracy. For instance, Sarwar et al. (1995) demonstrated that leveraging neighborhood-based approaches significantly enhances the performance of collaborative filtering systems by predicting user preferences based on the ratings of similar users. However, challenges such as noise and data sparsity often degrade model performance. To address these issues, recent models have incorporated techniques like contrastive learning and filtering components, which help reduce noise and improve the quality of recommendations by focusing on latent user-item embeddings. These advancements underscore the ongoing evolution in the design of collaborative filtering algorithms, aimed at enhancing the robustness and accuracy of recommendation systems. [20]

M. Dell’Amico and L. Capra on their research work, “SOFIA: Social filtering for robust recommendations,” Digital content production and distribution have transformed business models, creating an overwhelming supply to meet the global demand of millions of users. Since users cannot browse through vast numbers of items, filtering techniques have become essential for matching supply with demand. Traditional filtering methods identify trusted users based on either taste similarity ("I trust those who agree with me") or social ties ("I trust my friends and their trusted connections"). However, these approaches have limitations: taste similarity is vulnerable to manipulation, while social ties may overlook personal preferences. To address this, a new method called social filtering is proposed, which requires users to be both competent and well-intentioned to be trusted. The SOFIA algorithm, which embodies this approach, has demonstrated improved accuracy and robustness in real-world large-scale datasets. [21]

R. Sterritt on his research work, “Towards autonomic computing: effective event management,” Autonomic computing represents a groundbreaking approach in the design of computing systems, aiming to create systems that are self-managing, self-healing, self-protecting, and self-optimizing. This approach requires the integration of techniques from both software engineering and artificial intelligence. A key aspect of autonomic computing is event management, which encompasses a variety of event handling techniques, particularly in distributed systems. Intelligent event management approaches are exemplified in telecommunication systems, where the telecom survivable network architecture is analyzed to draw valuable lessons and identify potential challenges for autonomic computing in broader contexts. [22]

P. S. Hada, Yogesh, Bhupen, and Prince on their research work, “AI based Event Management Web Application,” The AI-Based Event Management web application is designed to streamline the process of discovering and registering for various events. This platform allows users to browse events by category, such as Techfests, workshops, webinars, conferences, and more, providing a centralized hub for event management. Registered users benefit from personalized recommendations based on their selected interests, ensuring that they receive suggestions for events that align with their preferences. The application supports new user registration and simplifies the event discovery process, making it easier for individuals to find and register for events that match their interests. The AI-driven recommendations enhance user experience by delivering relevant event suggestions, tailored to their specific preferences and past interactions with the platform. [23]

Chapter 3: System Analysis and Design

3.1 System Analysis

Feasibility, requirements and schedule of the proposed system are evaluated here.

3.1.1 Requirement Analysis

i. Functional Requirements

The system must provide following functionalities:

1. **User Functional Requirements:**
    - **User Registration**
      Users (organizers, attendees) must be able to create accounts and log in securely, ensuring personalized access and data protection.
    - **User Login**
      User should be able to login to app using their credentials which they used during sign up.
    - **User Profile**
      The user should be able to customize their user profile (i.e. their name and other information).
    - **Change Password**
      User should be able to change their password and set it to a new one.
    - **Browse Categories**
      User should be able to browse events according to their categories.
    - **Search Events**
      User should be able to search for events based on various criteria (e.g., category, date, location), utilizing advanced filtering and geolocation features to find relevant events easily.
    - **Event Creation and Management**
      Organizers should be able to create, edit, and manage event details including categories, dates, locations, participant estimates, and pricing, facilitating comprehensive event planning.

- **Ticketing and Payment Integration**

  The platform must support secure ticket purchasing and payment processing, enabling users to buy tickets for paid events seamlessly and safely.

2. **Admin Functional Requirements:**
   - **Admin Login**

     Admin should be able to login using their valid credentials.
   - **Manage users**

     Admin should be able to manage users (organization, individual) and review their transaction history. Admin should be able to add new users and assign roles.
   - **Manage events**

     Admin should be able to manage events. Admin should be able to add new events also should be able edit events information and delete events.

3. Use Case Diagram

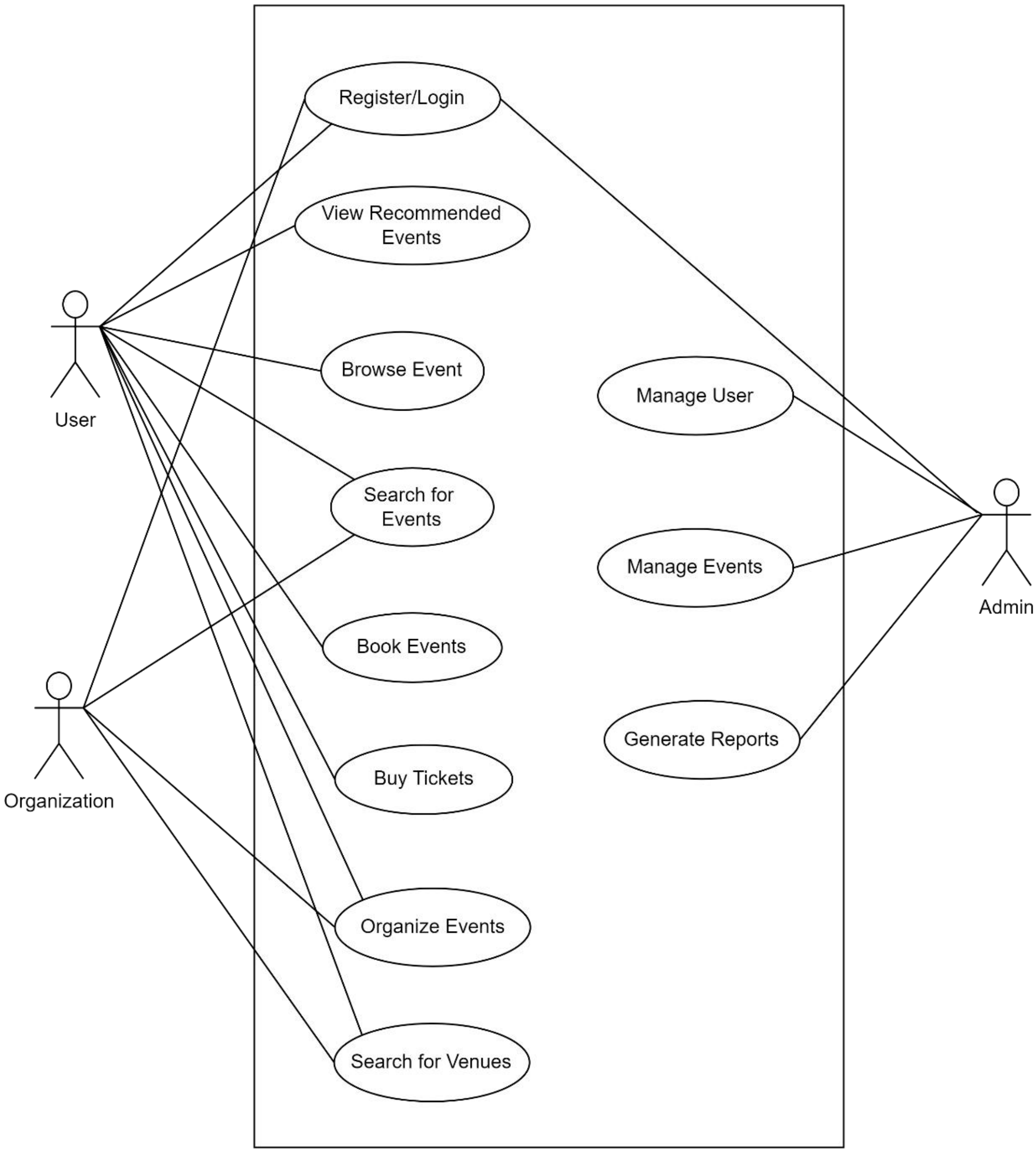


Figure 1: Use Case Diagram

Above use case diagram illustrates the main functionalities of the EventAura system, focusing on the interactions between the user and the system. Users can register, log in, and view recommended events. They can browse the website, search for events, book them, buy tickets, organize their own events, and search for venues. Administrators have additional capabilities to manage users, events, and generate reports, ensuring smooth operation and oversight of the platform.

ii. Non-Functional Requirement

Here are non-function requirements of the system:

1. Usability:

The system should be interactive and usable. It should provide a simple, interactive user interface which is easy to navigate and use such that the users won't have any troubles using the application.

2. Security:

EventAura should be designed in a way such that users are required to input a valid email and strong password for their account. The admin also has unique credentials such that the database is secure, and no attacks can happen to the data.

3. Performance:

The performance of the system is good. The response time is fast and without errors.

4. Maintainability:

EventAura can be modified and updated according to the growing needs. There is space for improvement and changes.

3.1.2 Feasibility Analysis

i. Technical Feasibility

The development stack can be easily scaled up or down based on demand. It has been determined that MySQL is a suitable database management system (DBMS) for the project due to its robustness, scalability, and open-source nature. The hardware requirements meet the stack requirements.

To deploy application, the only technical aspect needed for the server are mentioned below:

Operating System: Windows Server or any modern Linux distribution (e.g., Ubuntu, CentOS, Debian)

RAM: 4GB or higher

Disk Space: 20GB or higher

Processor: Dual-core processor or higher

Web Server Software: Apache or Nginx

Database: MySQL 5.7 or higher

Programming Languages: PHP 7.4 or higher

Front-end Technologies: HTML, CSS, JavaScript

ii. Financial Feasibility

The development of EventAura will require the use of free software and systems such as MySQL, git, postman, PHP and so on. This software are available free of cost and have extensive documentation and community support, thus no cost will be required for the development of this project.

iii. Operational Feasibility

EventAura demonstrates strong operational feasibility with its user-friendly interface, scalability to accommodate growth, accessibility across devices, robust support and maintenance framework, and ample technical resources ensuring smooth functionality and reliability.

iv. Schedule Feasibility

The time schedule for this project is 8 weeks. It is feasible to complete it within the given time period when the tasks are scheduled as follows:

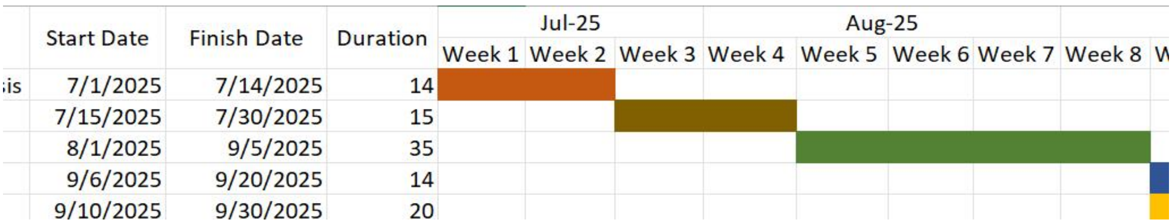


Figure 2: Gantt Chart

3.2 Data Modeling

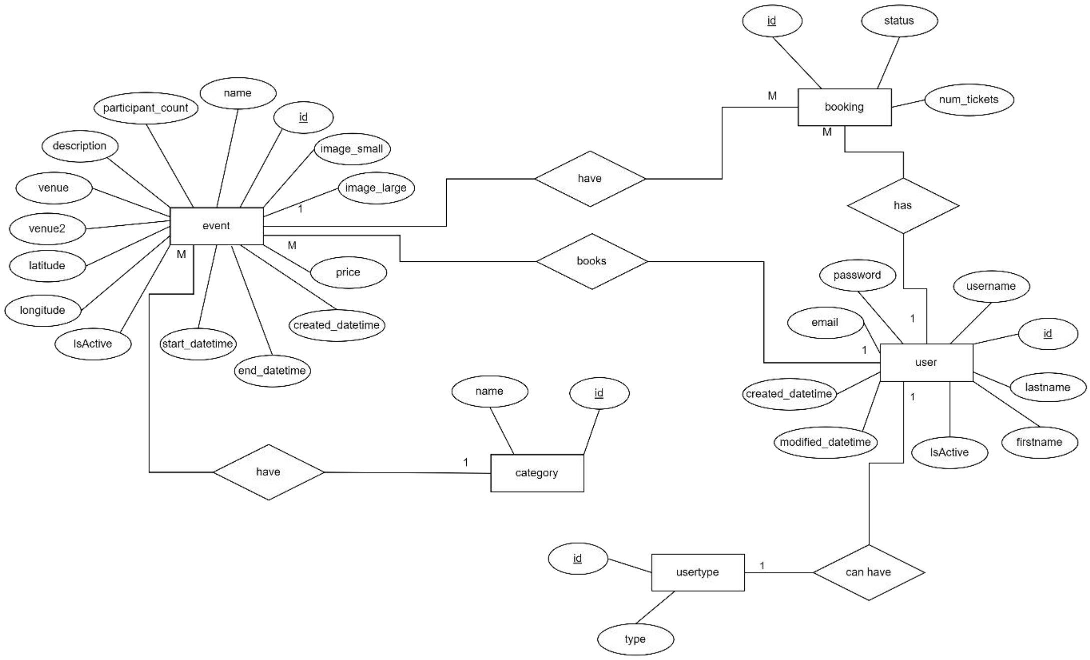


Figure 3: ER Diagram

3.3 Process Modeling

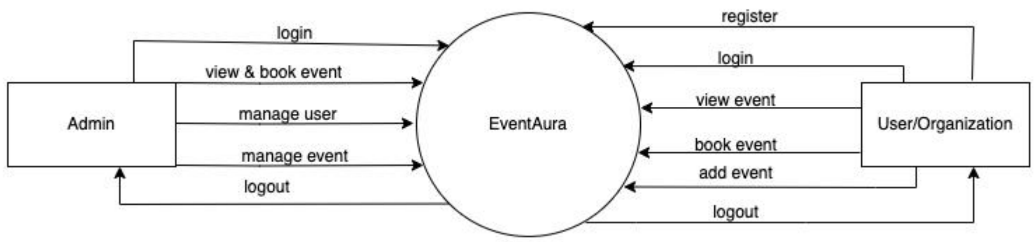


Figure 4: Context Level DFD of EventAura

The DFD of EventAura illustrates the flow of data and interactions between different users and the system. Users can register or login to view and book events. Administrators have additional privileges to manage users and events, including adding and managing events. The EventAura system acts as a central hub for event-related activities, facilitating interaction between users and administrators.

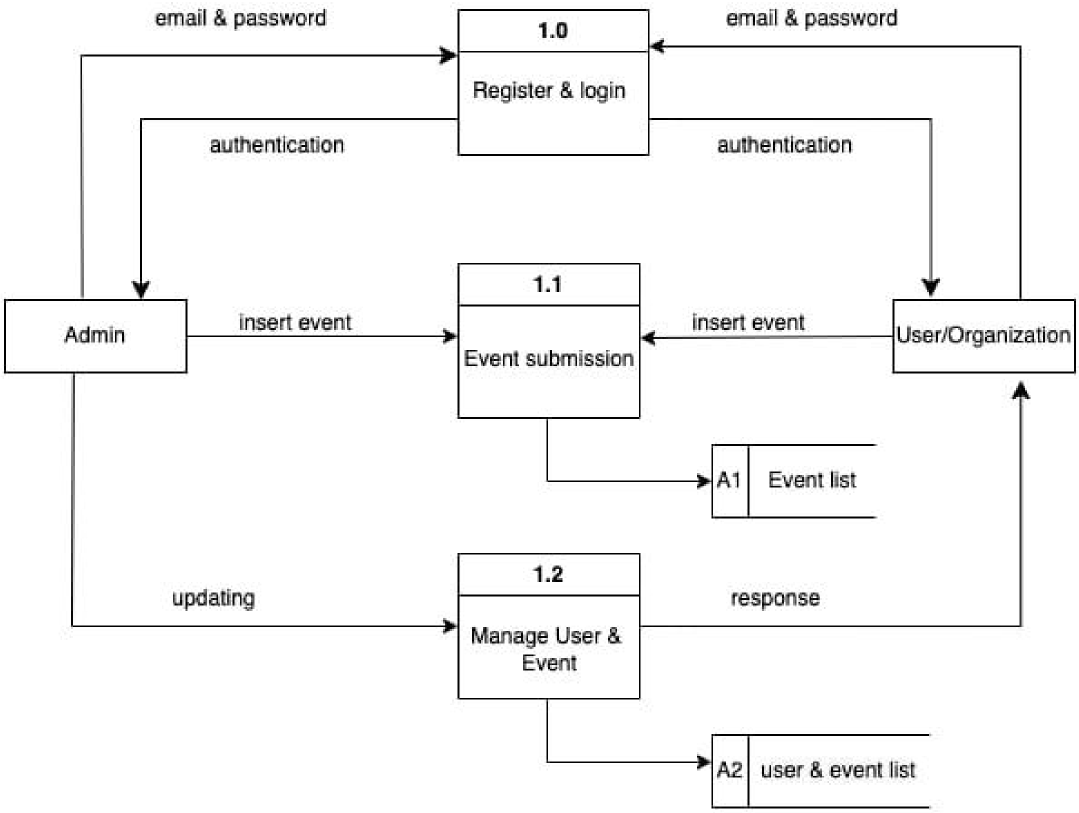


Figure 5: Level 1 DFD of EventAura

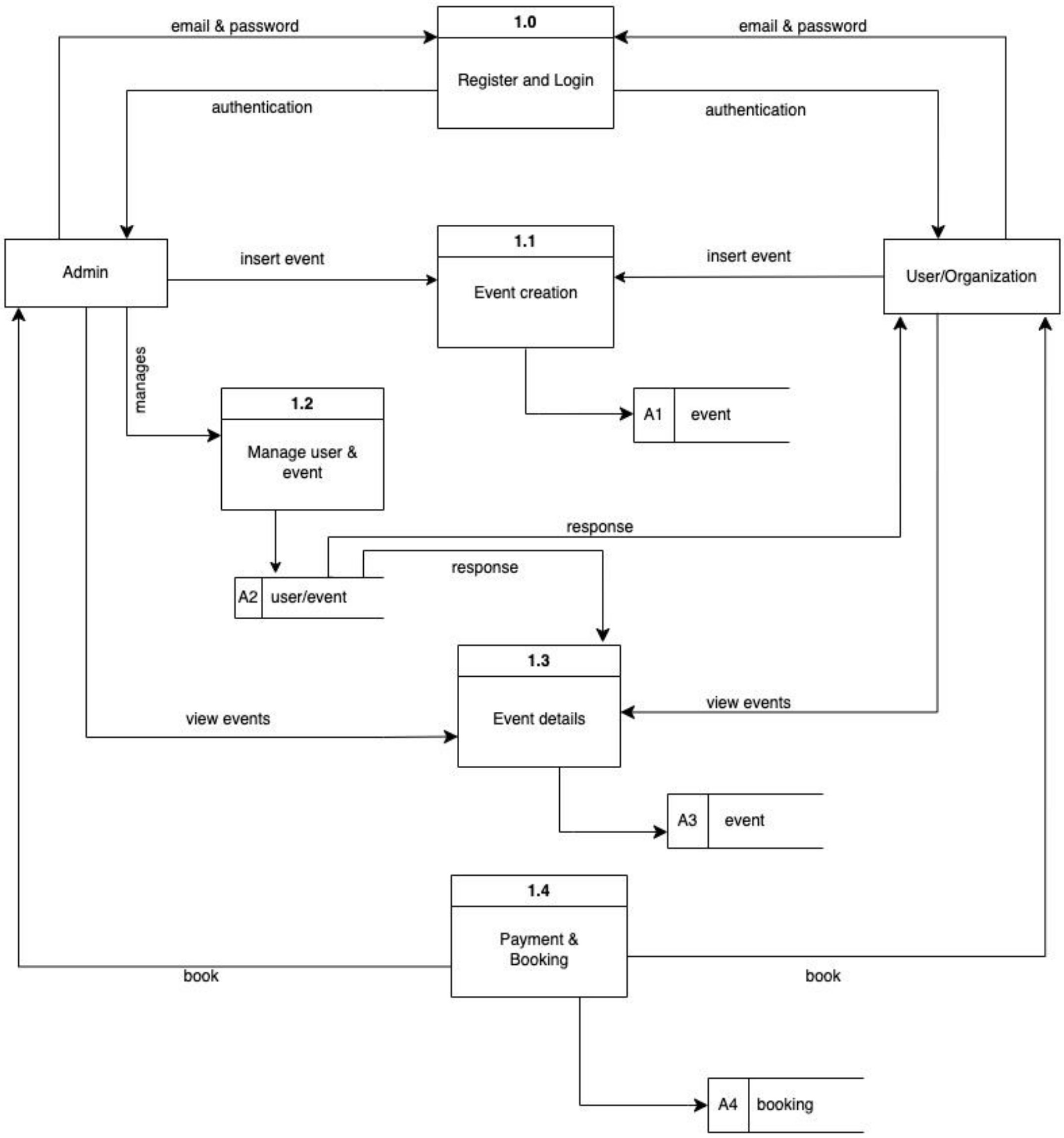


Figure 6: Level 2 DFD of EventAura

3.4 System Design

3.4.1 System Development Model

I will be using the waterfall model for the system. The reason I choose the waterfall model is because the requirements are fixed for this project so there will only be a single iteration.

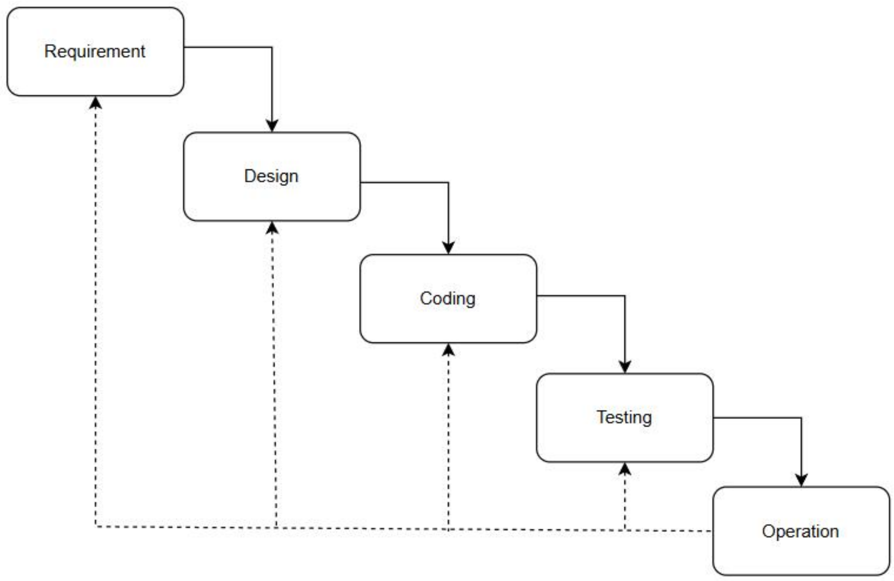


Figure 7: Waterfall Model

3.4.2 Architecture Design

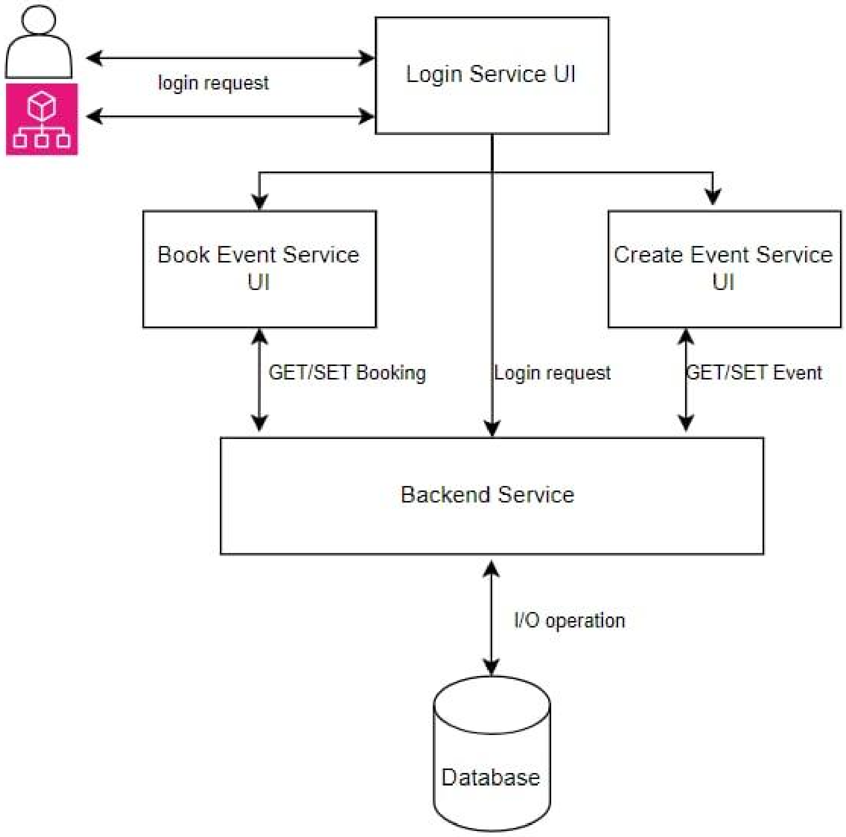


Figure 8: Architecture Design

This architecture diagram for EventAura shows how different components interact. Users access the system through various user interfaces (Login, Book Event, Create Event). These interfaces send requests to the central backend service, which handles the application's logic and communicates with the database to store or retrieve data. The backend service processes login requests, event bookings, and event creation, ensuring smooth data flow between the user interfaces and the database.

3.4.3 Database Schema Design

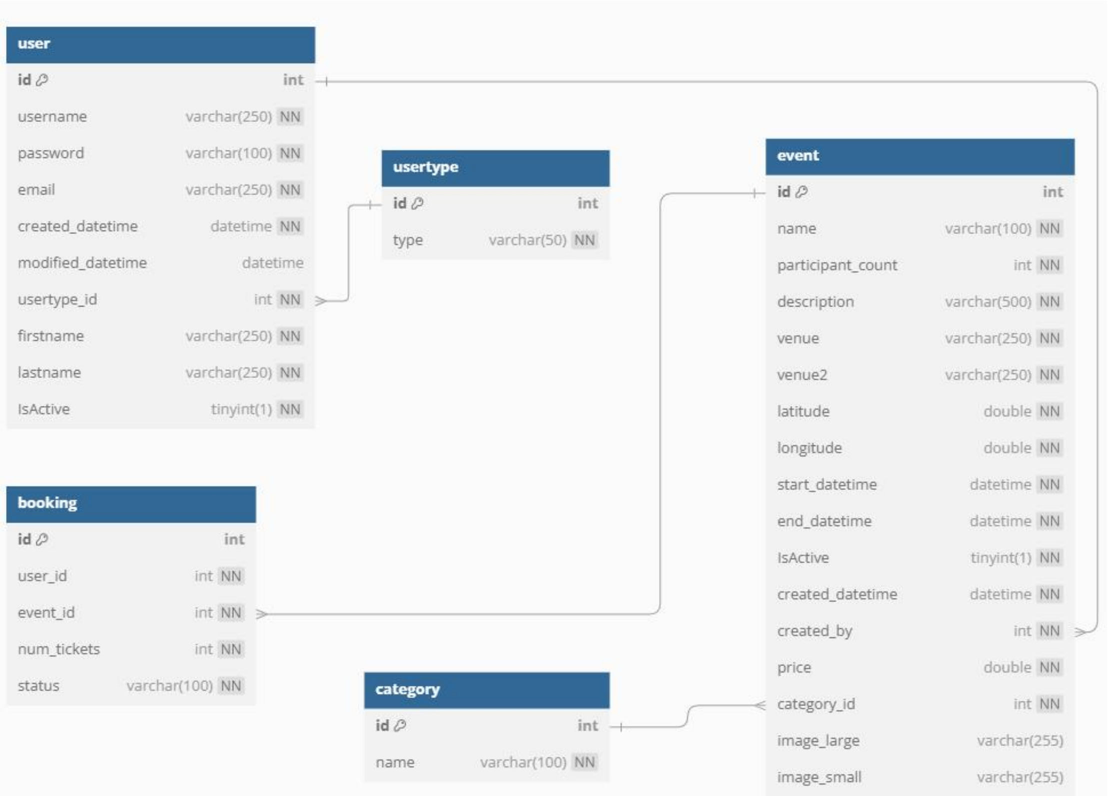


Figure 9: Schema Design

3.4.4 Interface Design (UI/UX)

Interface design is used to design how this project looks like and this design helps user to know how the system will look.

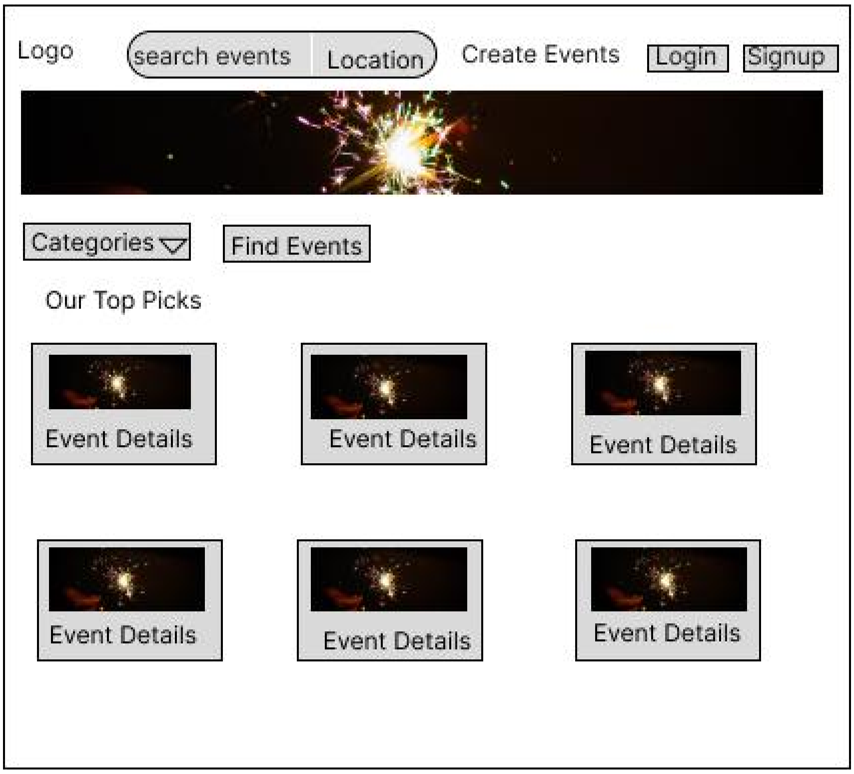


Figure 10: Home Page of EventAura

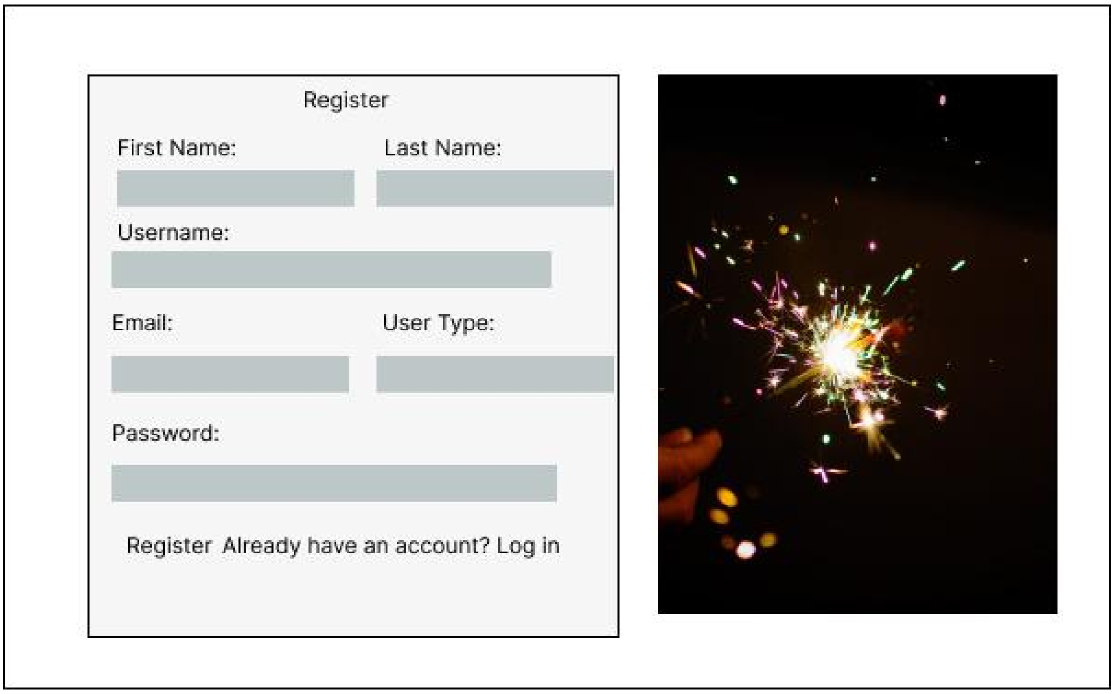


Figure 11: Signup Page of EventAura

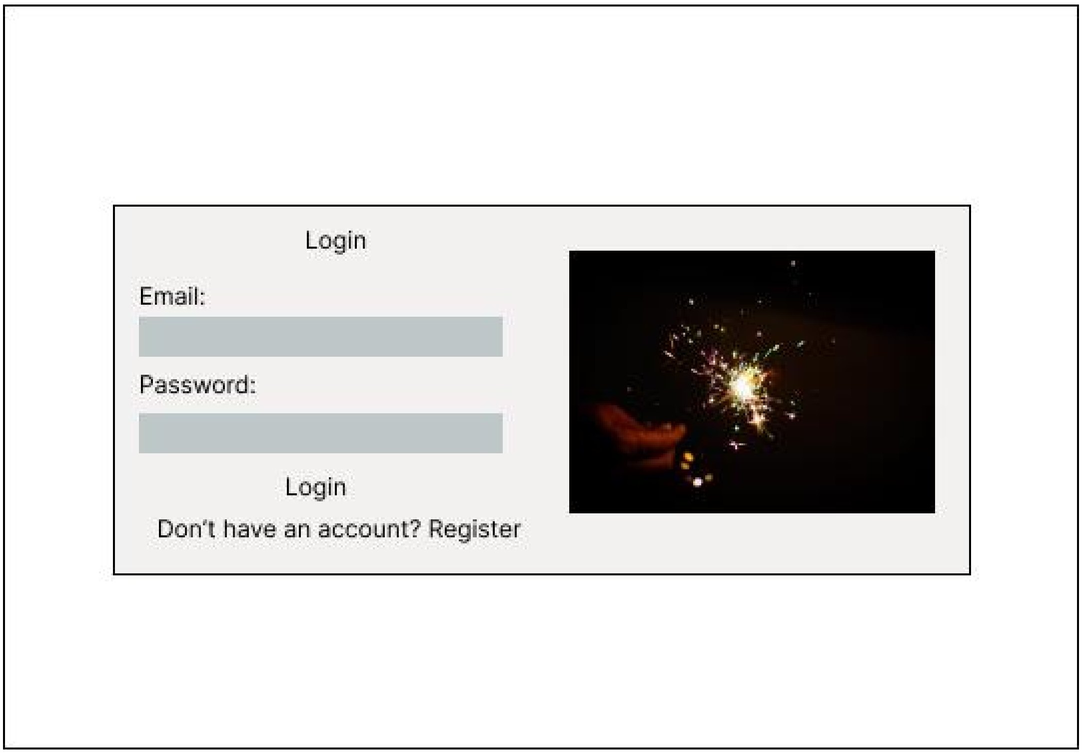


Figure 12: Login Page of EventAura

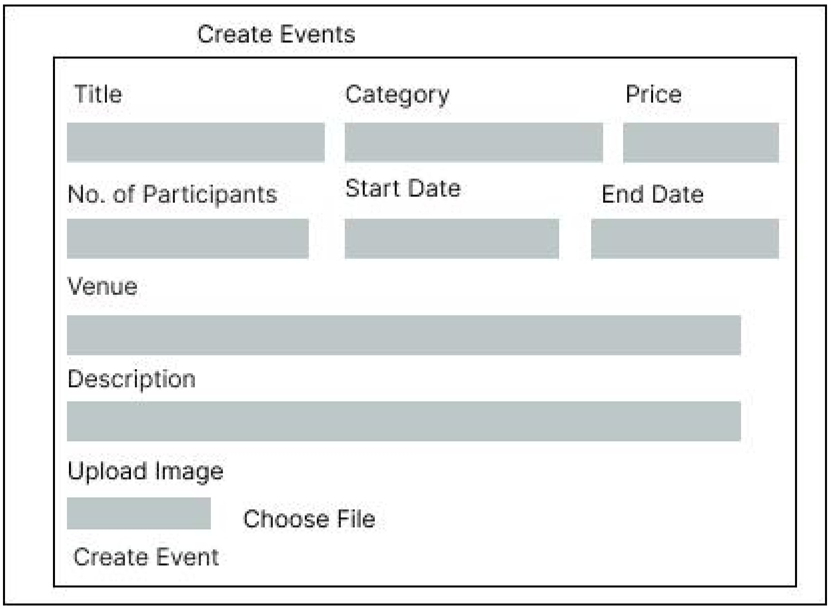


Figure 13: Create Event of EventAura

3.4.5 Physical DFD

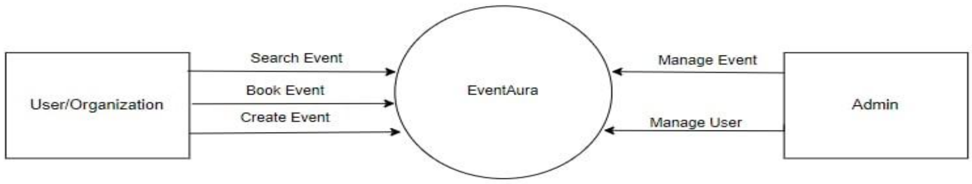


Figure 14: Physical DFD

This physical Data Flow Diagram (DFD) for EventAura illustrates the key interactions between the system and its users. On one side, users or organizations can interact with EventAura by searching for events, booking events, and creating new events. These actions are processed by the EventAura system. On the other side, the admin has control over managing events and users within the system. The diagram emphasizes the flow of data and tasks between the users/organizations, the EventAura system, and the admin, showing how each entity contributes to the overall functionality of the platform.

3.5 Description of Algorithm

The various algorithms used for this system are:

Recommendation Algorithms— Social Filtering

Social Filtering is a method used in recommendation systems that integrates both social network data and collaborative filtering techniques to improve the accuracy of recommendations. In EventAura, I'm employing a social filtering algorithm to recommend events to users based on the categories of events they have previously booked. The algorithm works by first identifying the categories of events a user has shown interest in (through past bookings) and then recommending new, unbooked events within those categories.

Mathematical Formula:

User-Category Association:

$$Cu = \{c1, c2, \ldots, cn\} \quad (1)$$

Where Cu represents the set of categories that user u has booked events in.

Event Recommendation:

For each event e in the set of all upcoming events E, the algorithm checks if the event's category c_e is in Cu and whether the user has already booked this event. This can be represented as:

$$Recommend(e) = \begin{cases} 1 \ if \ c_{ce} \in C_u \ and\ user\ has\ not\ booked\ e, \\ 0 \ otherwise. \end{cases} \quad (2)$$

Sorting by Start Date:

The recommended events are then sorted by their start date:

$$Sort(E_{rec}) = sort(E_{rec}, start\_datetime) \quad (3)$$

Where E_{rec} is the set of recommended events.

This approach tailors event suggestions to each user by focusing on their demonstrated interests while also ensuring they are introduced to relevant upcoming events they haven't yet booked.

```php
recommendation.php > findRecommendedEvents
<?php
function fetchBookedEvents($conn, $user_id) {
    $query = "
        SELECT e.id AS event_id, e.category_id
        FROM booking b
        JOIN event e ON b.event_id = e.id
        WHERE b.user_id = ? AND b.status = 'booked'
    ";
    $stmt = $conn->prepare($query);
    $stmt->bind_param('i', $user_id);
    $stmt->execute();
    $result = $stmt->get_result();
    $bookedEvents = [];
    while ($row = $result->fetch_assoc()) {
        $bookedEvents[] = $row;
    }
    $stmt->close();
    return $bookedEvents;
};
function getBookedCategoryIds($conn, $bookedEvents)
{
    $bookedCategories = [];
    foreach ($bookedEvents as $event) {
        $bookedCategories[] = $event['category_id'];
    }
    return $bookedCategories;
}
function fetchAllUpcomingEvents($conn){
    $query = "
    SELECT e.id, e.name, e.description, e.start_datetime, e.end_datetime, e.venue, e.category_id, e.image_small
    FROM event e
    WHERE e.start_datetime > NOW() AND e.IsActive = 1
    ";
    $result = $conn->query($query);
    $allEvents = [];
    while ($row = $result->fetch_assoc()) {
    $allEvents[] = $row;
    }
    return $allEvents;
}
function findRecommendedEvents($conn, $user_id){
    $bookedEvents = fetchBookedEvents($conn, $user_id);
    $allEvents = fetchAllUpcomingEvents($conn);
    $bookedCategories = getBookedCategoryIds($conn, $bookedEvents);
    $recommendedEvents = [];
    foreach ($allEvents as $event) {
        if (in_array($event['category_id'], $bookedCategories)) {
            $count=0;
            $query = "SELECT COUNT(*) FROM booking WHERE user_id = ? AND event_id = ?";
            $stmt = $conn->prepare($query);
            $stmt->bind_param('ii', $user_id, $event['id']);
            $stmt->execute();
            $stmt->bind_result($count);
            $stmt->fetch();
            $stmt->close();
            if ($count == 0) {
                $recommendedEvents[] = $event;
            }
        }
    }
    usort($recommendedEvents, function($a, $b) {
        return strtotime($a['start_datetime']) - strtotime($b['start_datetime']);
    });
    return $recommendedEvents;
}
```

Figure 15: Recommendation Implementation Code

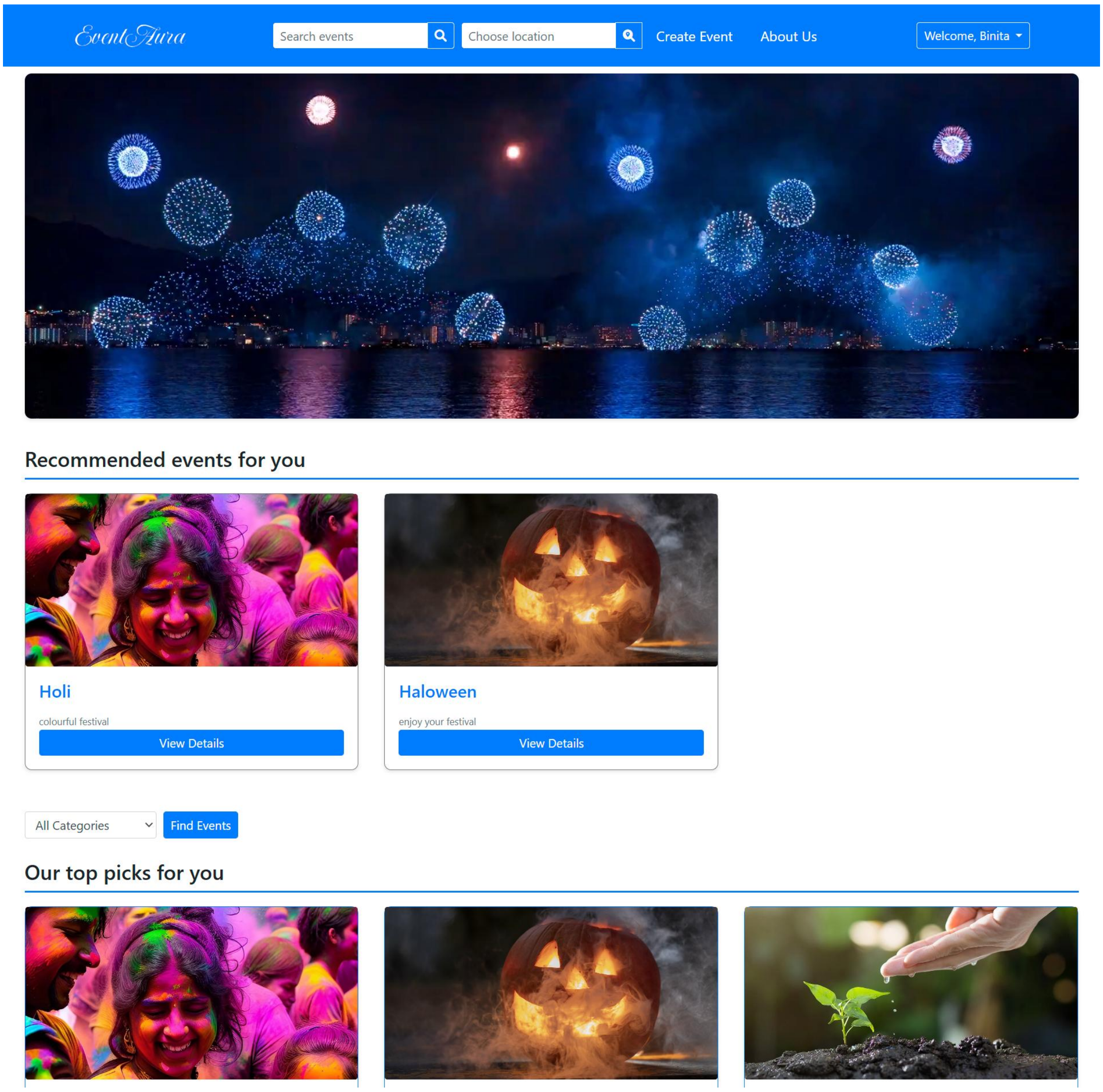


Figure 16: Recommendation Result

Haversine Algorithm:

The Haversine formula is used to calculate the shortest distance between two points on a sphere, given their latitudes and longitudes. In the context of EventAura the Haversine algorithm can be applied to determine the distance between a user's location and various event venues, facilitating geolocation-based search and recommendations.

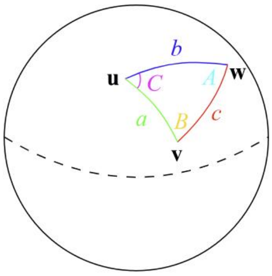


Figure 17: Haversine Formula

Mathematical Formula:

The Haversine formula calculates the great-circle distance d between two points (latitude and longitude) on the Earth's surface:

$$d = 2r.\operatorname{arcsin}\left(\sqrt{sin^2\left(\frac{\Delta lat}{2}\right) + \cos\left(lat_1\right).\cos\left(lat_2\right).sin^2\left(\frac{\Delta lon}{2}\right)}\right) \quad (4)$$

Where:

R is the radius of the Earth (mean radius = 6,371 km),

lat_1 and lat_2 are the latitudes of the two points,

lon_1 and lon_2 are the longitudes of the two points,

$$\Delta lat = lat_2 - lat_1 \quad (5)$$

$$\Delta lon = lon_2 - lon_1 \quad (6)$$

Description and Implementation:

Input Parameters: Requires latitude and longitude coordinates for the user's location and the venue's location.

Calculate Differences: Compute the differences in latitude and longitude between the user and venue locations.

Apply Trigonometric Functions: Use trigonometric functions to compute the Haversine formula, determining the distance in kilometers or miles.

Recommendation Use: Use the computed distances to filter and recommend events or venues within a specified radius from the user's location.

The Haversine algorithm enables EventAura to provide users with accurate distance-based recommendations, enhancing their ability to find nearby events and venues of interest. This capability improves user engagement and satisfaction by offering relevant and geographically suitable options through the platform.

searchbylocation.php > ...

```php
<?php
session_start();
require 'includes/database.php';
$conn = getDB();

$latitude = isset($_POST['latitude']) ? $_POST['latitude'] : '';
$longitude = isset($_POST['longitude']) ? $_POST['longitude'] : '';
if (!empty($latitude) && !empty($longitude)) {
    // Radius in kilometers
    $radius = 20;
    // Get today's date in the format that matches the database
    $today = date('Y-m-d H:i:s');

    // Prepare the SQL query with Haversine formula
    $sql = "
        SELECT *, (
            6371 * acos(
                cos(radians(?)) * cos(radians(latitude)) * cos(radians(longitude) - radians(?)) +
                sin(radians(?)) * sin(radians(latitude))
            )
        ) AS distance
        FROM event
        HAVING distance < ?
        AND IsActive = 1
        AND start_datetime IS NOT NULL
        AND end_datetime IS NOT NULL
        AND start_datetime >= NOW()
        ORDER BY start_datetime DESC
    ";
    try {
        // Prepare and execute the query
        $stmt = $conn->prepare($sql);
        if (!$stmt) {
            throw new Exception("Preparation failed: " . $conn->error);
        }

        $stmt->bind_param('dddd', $latitude, $longitude, $latitude, $radius);

        $stmt->execute();
        $result = $stmt->get_result();

        if (!$result) {
            throw new Exception("Error fetching events: " . $conn->error);
        }

        $events = $result->fetch_all(MYSQLI_ASSOC);

    } catch (Exception $e) {
        // Handle any errors
        die($e->getMessage());
    }
} else {
    // If latitude and longitude are not provided, get all events
    $sql = "SELECT * FROM event WHERE IsActive = 1 ORDER BY start_datetime DESC";

    try {
        // Prepare and execute the query
        $result = $conn->query($sql);

        if (!$result) {
            throw new Exception("Error fetching events: " . $conn->error);
        }
```

```php
            $events = $result->fetch_all(MYSQLI_ASSOC);

        } catch (Exception $e) {
            // Handle any errors
            die($e->getMessage());
        }
    }

    // Pass search parameters to the next page
    $_SESSION['search_events'] = [
        'events' => $events
    ];

    // Redirect to the search results page
    header("Location: searchresults.php");
    exit;
    ?>
```

Figure 18: Haversine Formula Implementation code

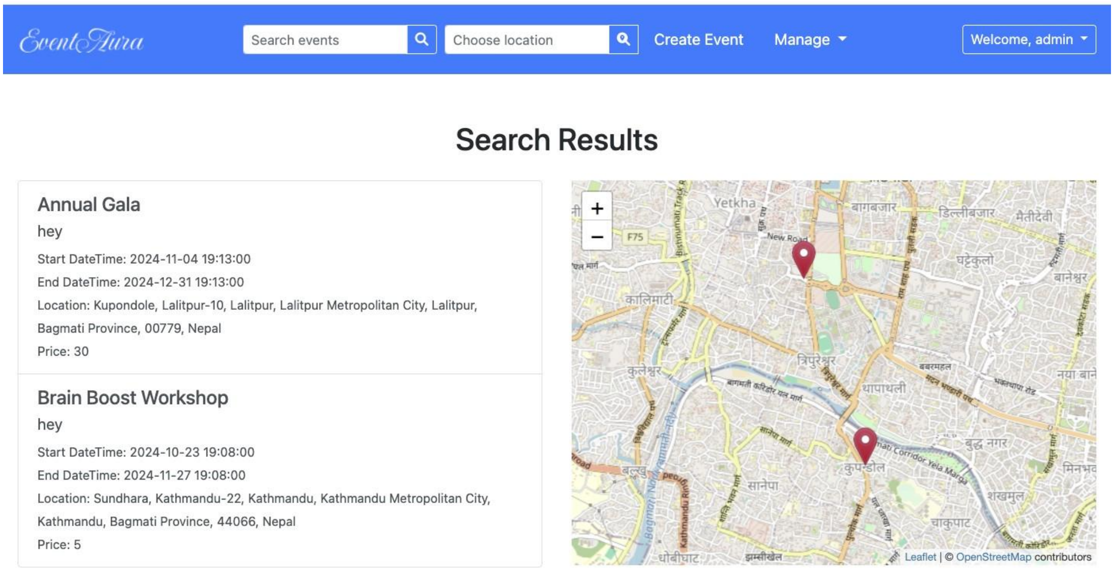


Figure 19: Location Search Result

Chapter 4: Implementation and Testing

4.1 Implementation

4.1.1 Tools Used

Table 1: Tools and Technologies

Tools	**Purpose**
HTML/CSS	Website design and frontend development
JS	Frontend development and responsiveness
MySQL	Database system
PHP	Backend development
Apache	Web Server Software
Figma	Design Tool
Draw.io	Diagram Tool

- **HTML/CSS:**
  HTML is a markup language for the web that defines the structure of web page. EventAura is developed using the HTML which helped in developing the user interfaces. CSS (Cascading Style Sheets) is a fundamental technology used in web development, including the development of the EventAura. It provides the means to control the visual aspects and layout of the EventAura.
- **VS Code:**
  All of the code of EventAura is written in Visual Studio Code (VS Code) which is a source-code editor made by Microsoft.

- **JavaScript:**

  JavaScript is a programming language which is used to build the logic while manipulating the data in EventAura. It enhances the user experience by adding interactivity and dynamic behavior to the website.

- **PHP:**

  PHP (Hypertext Preprocessor) is a server-side scripting language which is used to build the logic in backend of EventAura. PHP handles user requests, processes data, and interacts with the MySQL database to provide dynamic content to users.

- **MySQL:**

  MySQL is open source database management system that runs on server. It is used to store collected data of EventAura. It stores and manages critical data such as event schedules, bookings, and reviews.

- **Apache:**

  Apache is the actual web server application that processes and delivers web content to a computer. It is responsible for hosting and delivering web content to user in EventAura. Apache handles incoming requests, manages user interactions with the software's web interface, and ensures secure and reliable communication.

- **Figma:**

  Figma is used in EventAura for designing the user interface (UI) and creating interactive prototypes.

- **Draw.io:**

  Draw.io is used in EventAura for creating diagrams like flowcharts, data flow diagrams (DFDs), and architectural designs.

4.1.2 Implementation of modules

User Module

Login Module:

This screen contains form with two fields email and password and login button to submit the form. If valid user credentials is provided, then the user is redirected to the home screen. If either of the field is incorrect or left blank, an error message is displayed, and user should fill the information again.

Sign-up Module:

This screen contains form with fields to retrieve information (email, address, password) and sign-up button to submit the form. After submitting the form, if the user inserts valid data, the user will be signed up and be redirected to the log in page. If email is already in use or any field does not satisfy form requirement as left blank or weak password, an error message should be displayed.

Home Module:

The header includes search bar, create events and Login/Signup. The body section contains banners, events, services and an about us section.

Create Event Module:

This screen consists of form for creating events with its detail information. All the fields need to be filled to create events. The user or event creator can add details like event title, venue, start and end date, no. of participants, price, event description, also images related to events.

Search Module:

The search module has a global search function. The user can use the search bar to search for specific events by name of events and user also can search for events by their location. The search bar searches for all the events by their name or location entered.

Admin Module

The admin modules can do all the internal functionalities of the system like managing users i.e. editing user information and deleting users, managing events i.e. editing events information and deleting events also admin can create new events.

4.2 Testing

EventAura undergoes a comprehensive testing process that includes unit testing and system testing. Each phase ensures the functionality, performance, and user experience align with the specified requirements. Automated and manual testing methods are employed to identify and rectify issues, ensuring a robust and reliable fitness club management system.

4.2.1 Test Cases for Unit Testing

S. N	Description	Expected Result	Actual Result	Test Result
1	This screen contains form with two fields email and password and submit button to submit the form.	Home Screen of "EventAura" should be open after login.	Home Screen is opened successfully.	**PASS**
2	User enters a wrong email. Email: abcd Password: 1234	Display message User not found.	As expected,	**PASS**
3	User enters a wrong password. Email: shweta12@gmail.com Password: 123456	Display message Invalid credentials.	As expected,	**PASS**
4	User enters correct email and password. Email: shweta12@gmail.com Password: 12345	User logs in successfully.	As expected,	**PASS**

Table 2: Test Result for Login Module

Table 3: Test Result for Signup Module

S. N	Description	Expected Result	Actual Result	Test Result
1	User selects already existing email and other details. Full Name: Shweta Address: balaju Email: shweta12@gmail.com Password: 12345	Display message User already exist.	As expected,	**PASS**
2	User forgets to enter a particular required field. Full Name: Shweta Address: Email: shweta12@gmail.com Password: 12345	Display message One or more fields are empty.	As expected,	**PASS**
3	User enters the invalid email format. Full Name: Shweta Address: balaju Email: shweta Password: 12345	Display message Invalid email.	As expected,	**PASS**
4	User enters all the details successfully. Full Name: Jyoti Address: balaju Email: jyoti@gmail.com Password: 1234	User account created.	As expected,	**PASS**

Table 4: Test Result for Create Event Module

S. N	Description	Expected Result	Actual Result	Test Result
1	This screen consists of form with fields to retrieve event information and Create Event button to create event. If every field are filled.	User should be redirected to home page.	User was redirected to home page.	**PASS**
2	If any of the fields are left blank.	Please fill out this field message should be displayed.	Please fill out this field message was displayed.	**PASS**

Table 5: Test Result for Search Event Module

S. N	Description	Expected Result	Actual Result	Test Result
1	The search event function allows user to search for specific events. Search: Holi	Only the events whose name match the search input should be displayed. Events related to “holi” should be displayed.	Events related to “holi” were displayed.	**PASS**
2	If the input doesn’t match any existing events. Search: ababxy	“No events found” message should be displayed.	“No events found” message was displayed.	**PASS**

Table 6: Test Result for Search by Location Module

S. N	Description	Expected Result	Actual Result	Test Result
1	The search by location function allows user to search for specific events using location. Search: Kathmandu	Only the events whose name match the search input should be displayed. Events located in “Kathmandu” should be displayed.	Events located in “Kathmandu” were displayed.	**PASS**

2	If the input doesn't match any existing events. Search: ababxy	Default events should be displayed.	Default events was displayed.	**PASS**

Table 7: Test Result for Admin Module

S. N	Description	Expected Result	Actual Result	Test Result
1	Admin login page is displayed first for security.	Admin should be able to login and view the admin panel.	Admin was able login and view admin panel.	**PASS**
2	If the admin enters a wrong email. Email: admin123 Password: Admin	Display message User not found.	As expected,	**PASS**
3	If admin enters wrong password. Email: Admin Password: admin12	Display message Invalid credentials.	As expected,	**PASS**
4	The admin panel consist of all the existing events and user details, manage user and manage events.	Admin should be able to view all events and users and interact with various functions.	Admin was able to view all events and users and interact with various functions.	**PASS**
5	Manage Events function consists of a form that inputs the detail of events.	Admin should be able to edit and delete the existing events and also create events.	Admin wasable to edit and delete the existing events and also create events.	**PASS**
6	Manage User function consists of a form that inputs the detail of user.	Admin should be able to edit and delete the existing users.	Admin was able to edit and delete existing user.	**PASS**

4.2.2 Test Cases for System Testing

In the system testing phase of EventAura, the entire event management system is evaluated as a whole to ensure all integrated components work seamlessly.

Table 8: Test Result of System Testing for User

S. N	Test Case	Expected Result	Actual Result	Test Result
1	To check if user account is created or not.	User account should be created.	User account was created.	**PASS**
2	To check if the login module works.	User should be redirected to home page.	User redirected to home page.	**PASS**
3	To check if the home page is responsive.	Home page components should be responsive.	Home page was responsive.	
4	To check if create event module works.	User should be able to create events.	User was able to create events.	**PASS**
5	To check if search event module works.	User should be able to search events.	User was able to search events.	**PASS**
6	To check if search by location module works.	User should be able to search events by their location.	User was able to search events by their location.	**PASS**

Table 9: Test Result of System Testing for Admin

S. N	Test Case	Expected Result	Actual Result	Test Result
1	To check if admin account is created or not.	Admin account should be created.	Admin account was created.	**PASS**
2	To check if the login module works.	Admin should be redirected to home page.	Admin redirected to home page.	**PASS**
3	To check if manage events function works.	Admin should be able to edit and delete events and also create events.	Admin was able to edit and delete events and also create events.	**PASS**
4	To check if manage user function works.	Admin should be able to edit and delete users.	Admin was able to edit and delete users.	**PASS**

Chapter 5: Conclusion And Future Recommendations

5.1 Outcomes

The EventAura project aims to deliver a comprehensive event management platform tailored to the needs of both event organizers and attendees. Upon completion, the platform will enable users to seamlessly discover, register for, and participate in various events. Event organizers will benefit from a user-friendly interface that allows them to create, manage, and promote their events, while attendees will enjoy a personalized experience through event recommendations based on their interests and preferences.

Additionally, the platform will streamline the entire event lifecycle, from venue selection and ticket sales to participant engagement and feedback collection. EventAura will contribute to enhancing community interaction by making it easier for individuals and organizations to connect through shared interests and activities. The project's successful implementation will not only provide a robust technical solution for event management but also foster a more vibrant and connected community by simplifying the way people engage with events.

5.2 Conclusion

The completion of the EventAura project marks a significant milestone in the field of event management technology. This platform has been designed to address the diverse needs of both event organizers and participants, offering a streamlined and efficient way to manage and engage with events. By providing a robust and user-friendly interface, EventAura simplifies the complex processes of event creation, promotion, and participation, making it accessible to a wide range of users.

The platform's capabilities, including personalized event recommendations, secure ticketing, and venue management, not only enhance user experience but also contribute to a more connected and active community. EventAura stands as a testament to the power of technology in bringing people together, fostering social interaction, and promoting community engagement. The project's success sets a strong foundation for future innovations and improvements in the event management industry, paving the way for more inclusive and interactive experiences.

5.3 Future Recommendation

- Mobile App Development
- Social Media Integration
- Event Sponsorship and Advertising
- AI-driven Recommendation Engine
- Multi-language Support
- Improve portal UI

Appendices (Screenshots)

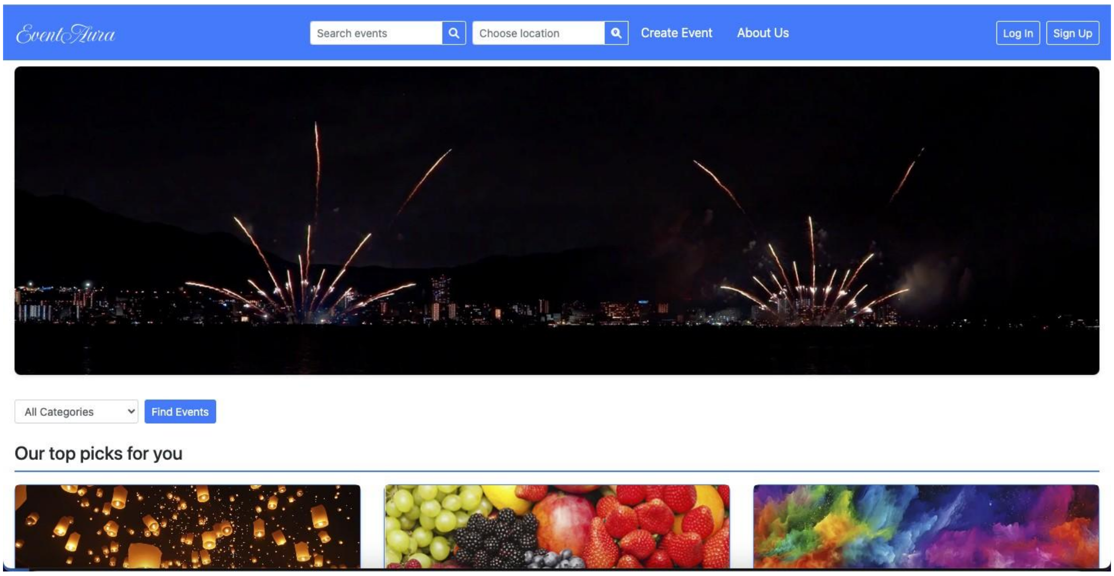


Figure 20: Home Page

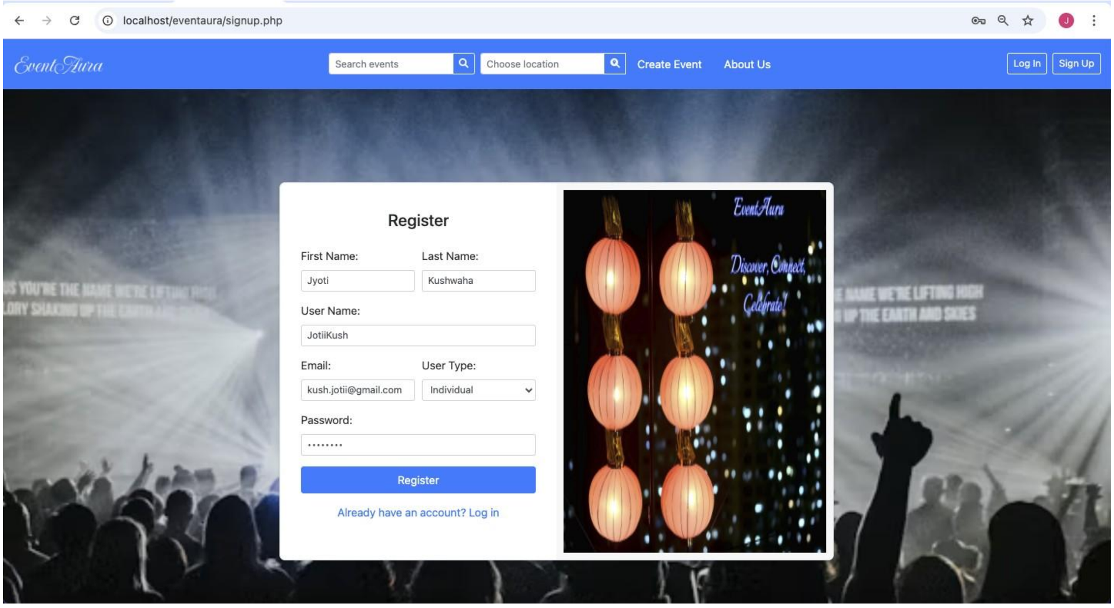


Figure 21 : Signup Page

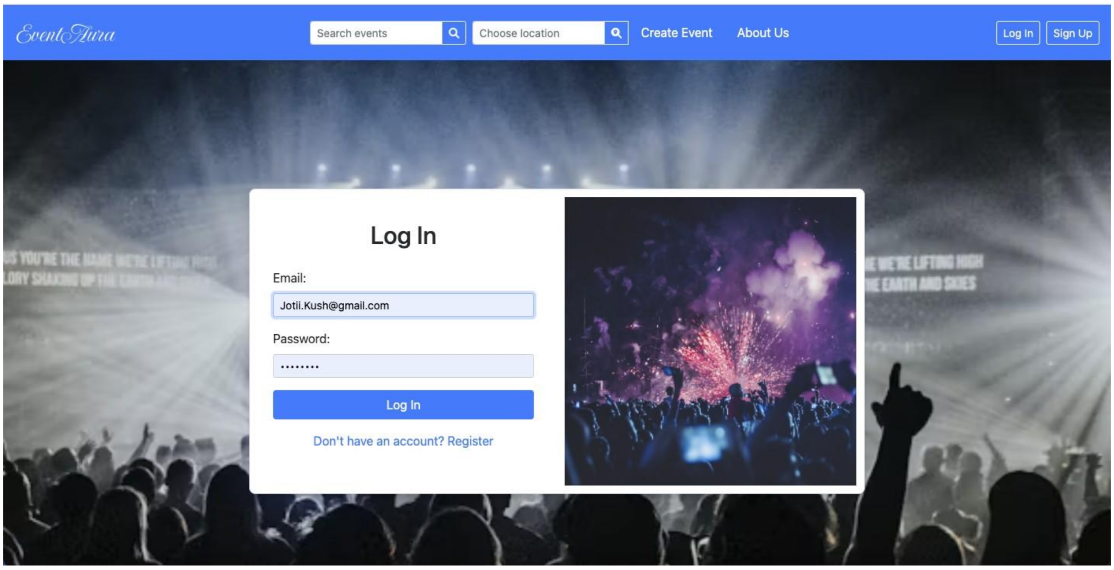


Figure 22: Login Page

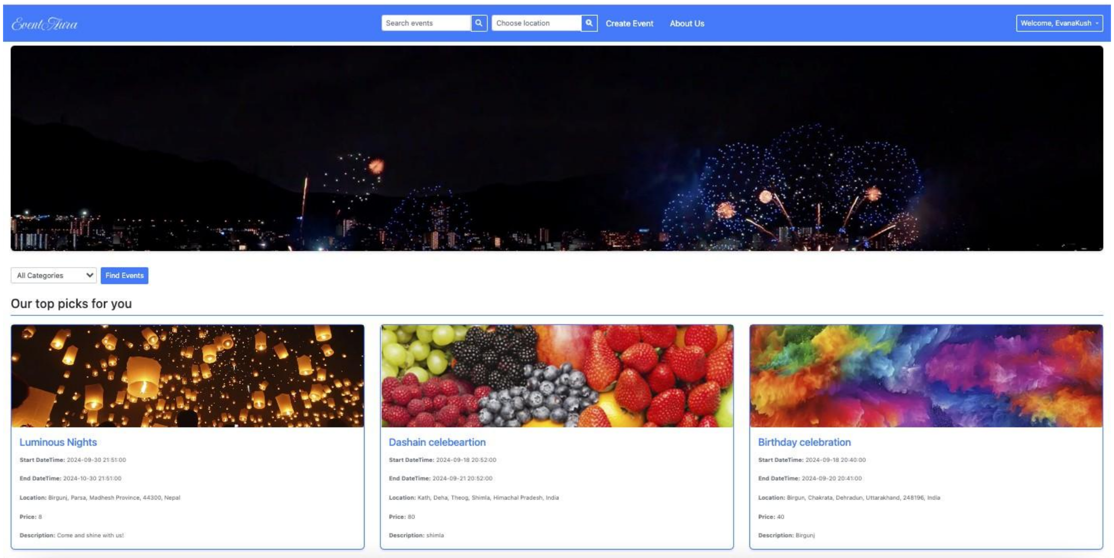


Figure 23: Home Page After Login

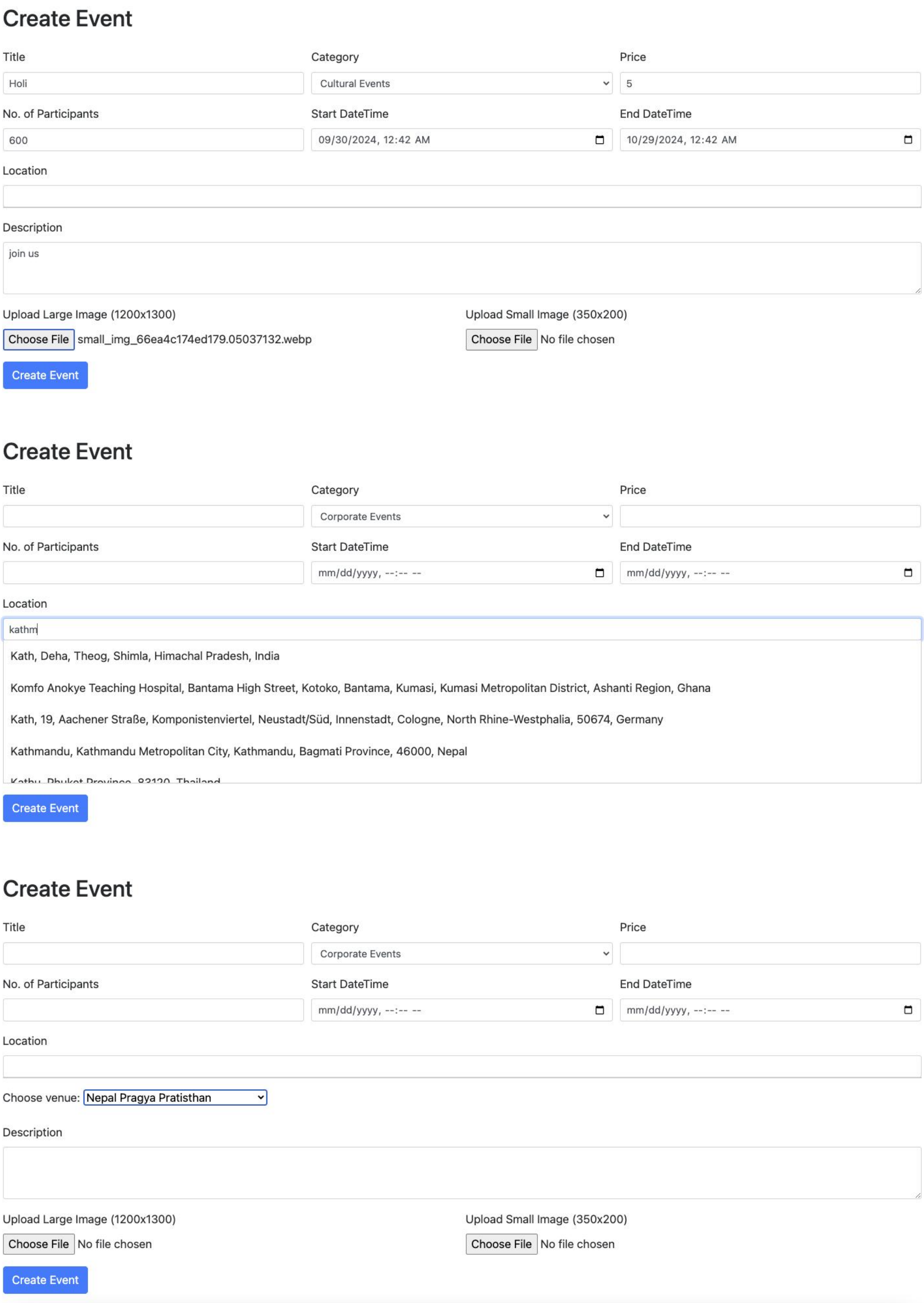


Figure 24: Create Event Page

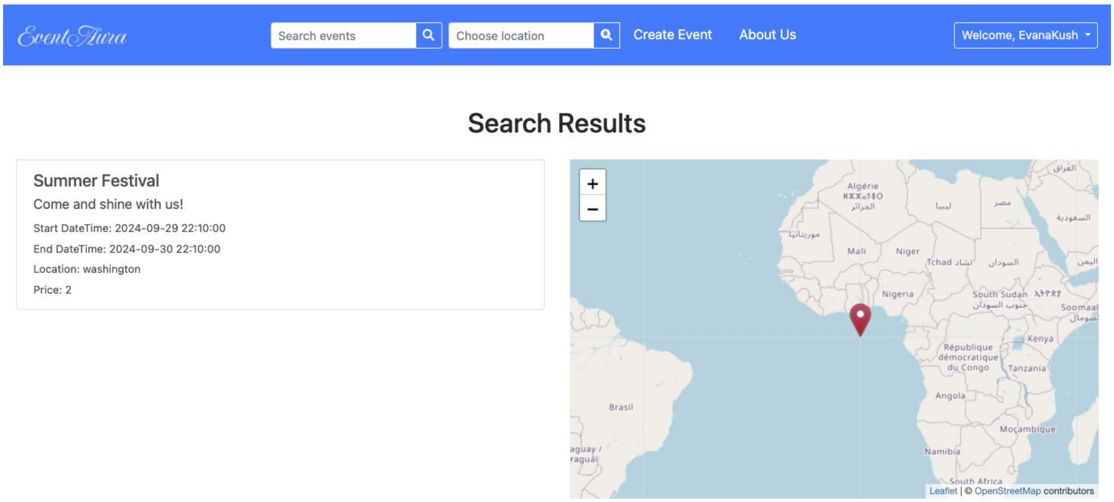


Figure 25: Search by Text Page

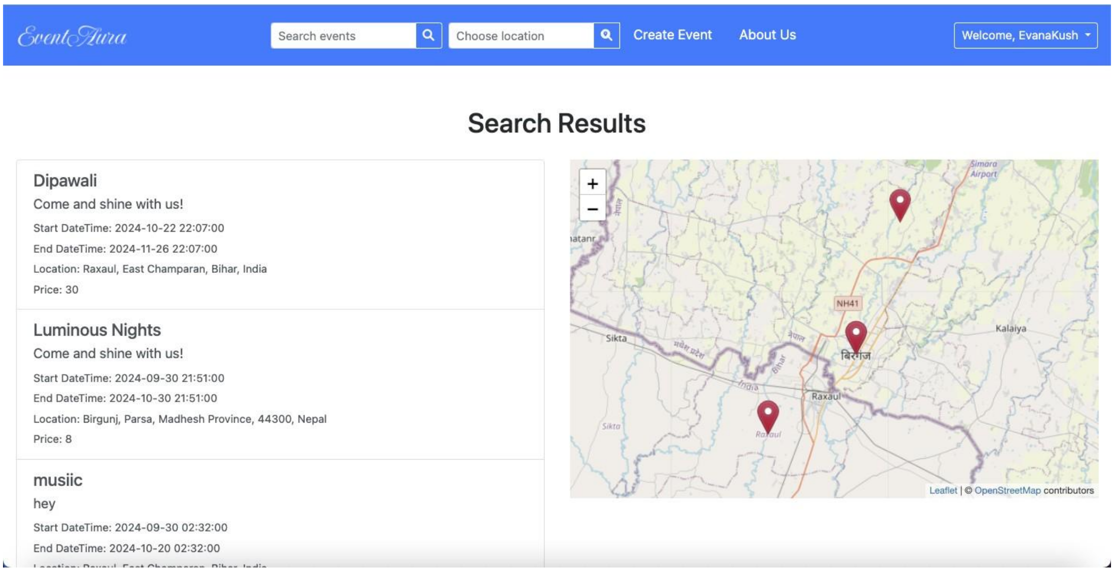


Figure 26: Search by Location Page

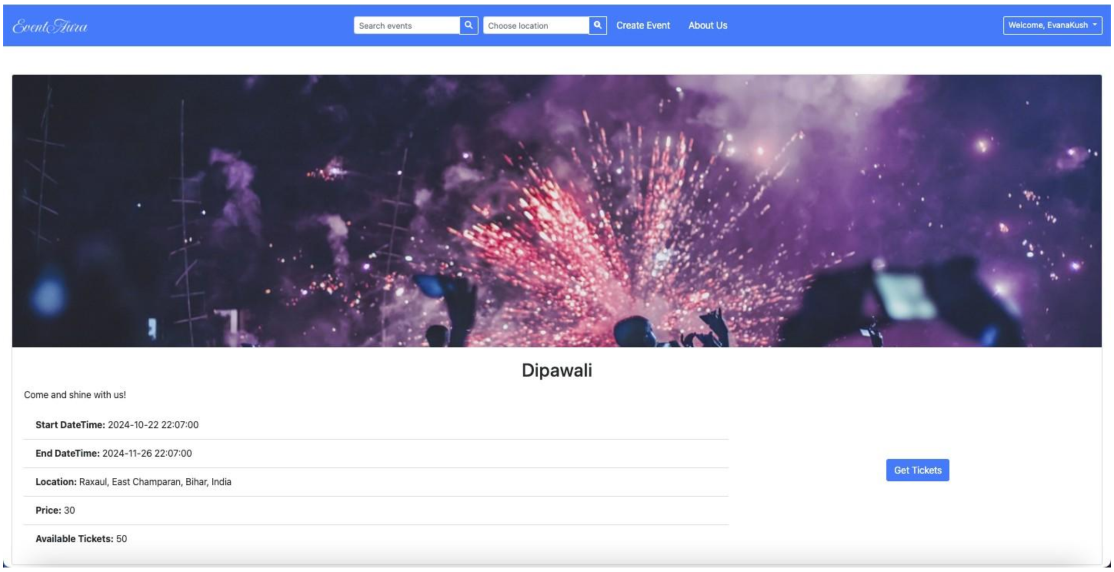


Figure 27: Event Detail Page

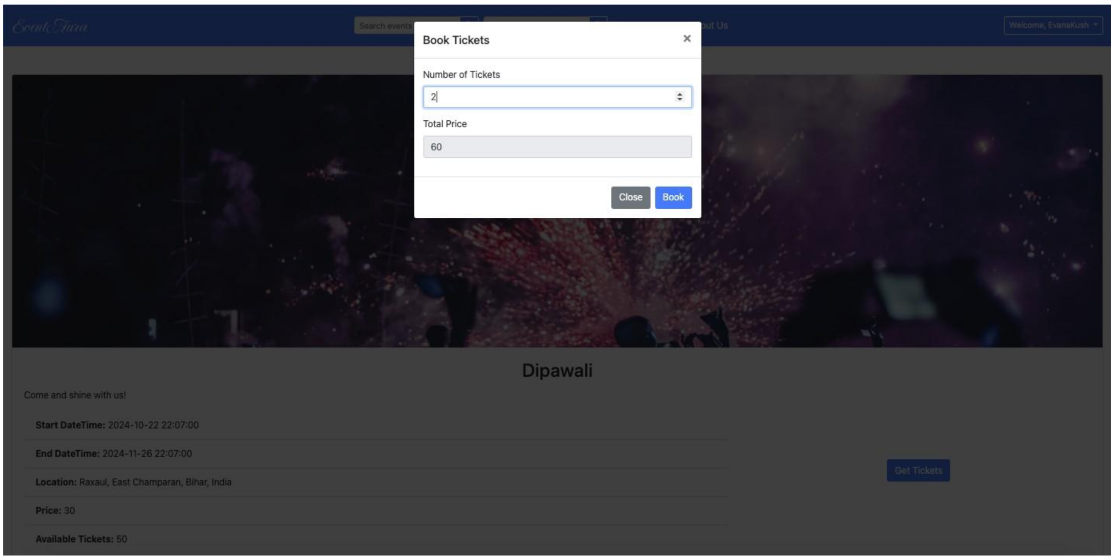


Figure 28: Book Ticket Popup

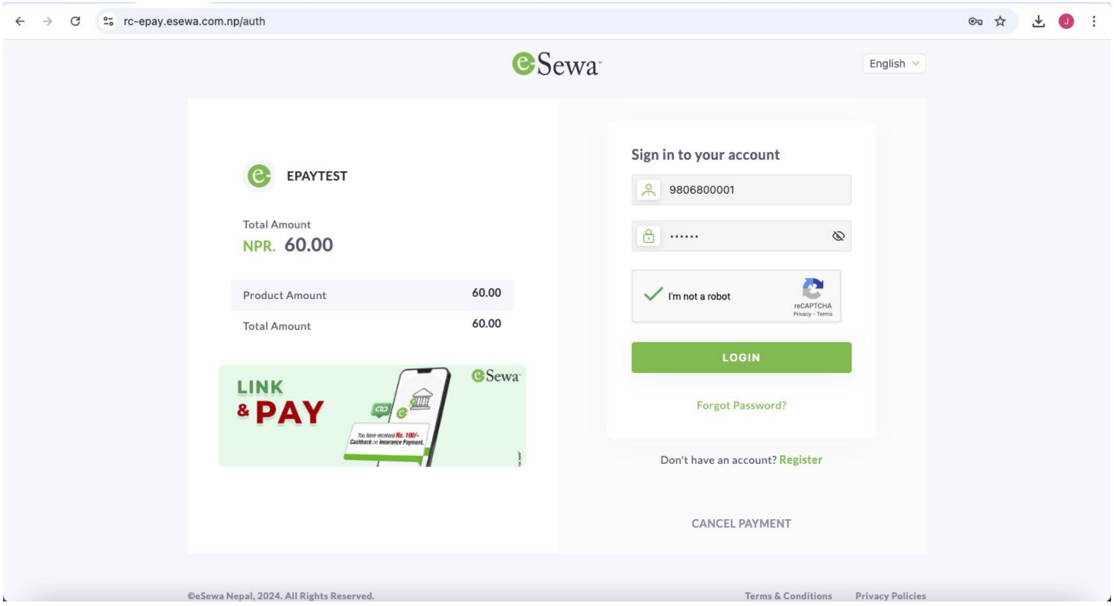


Figure 29: Payment Gateway Page

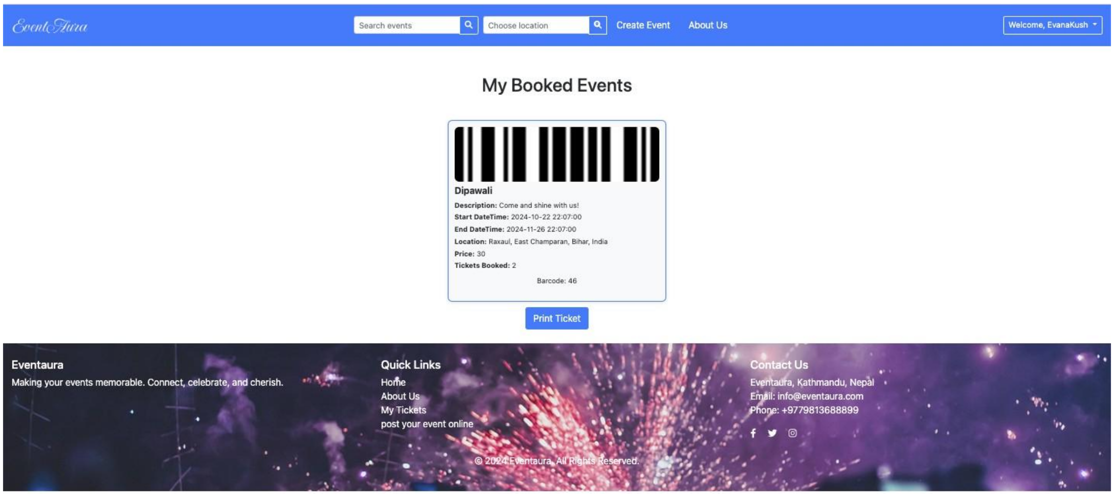


Figure 30: Booked Events Detail Page

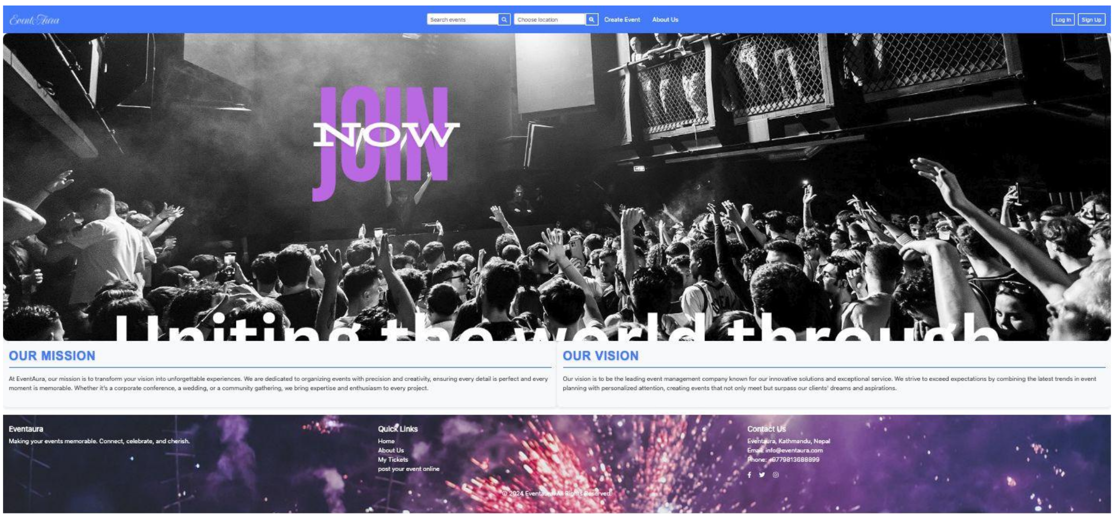


Figure 31: About Us Page

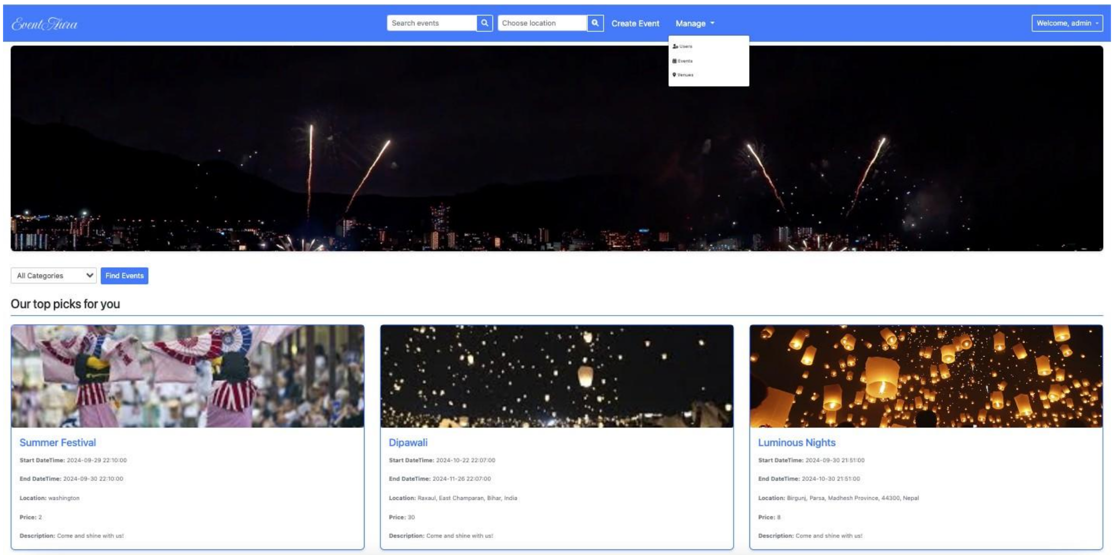
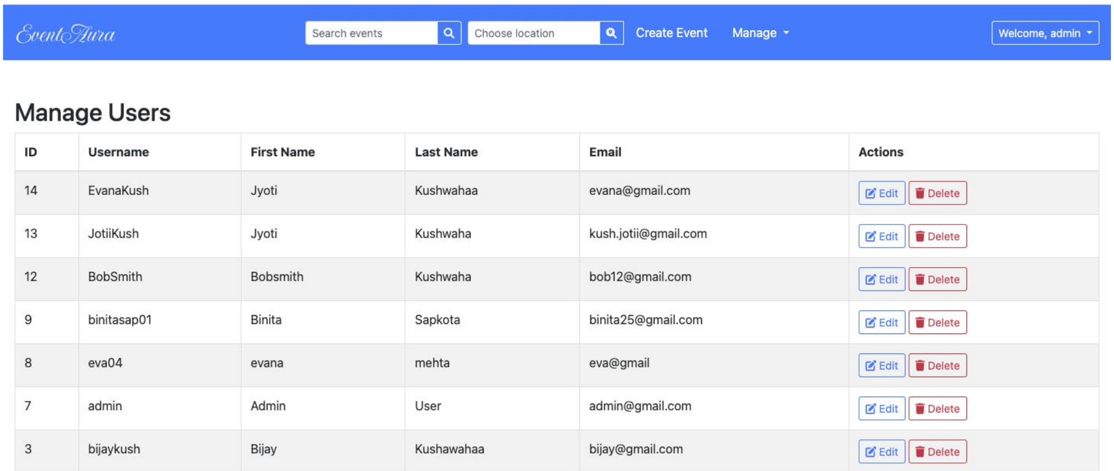
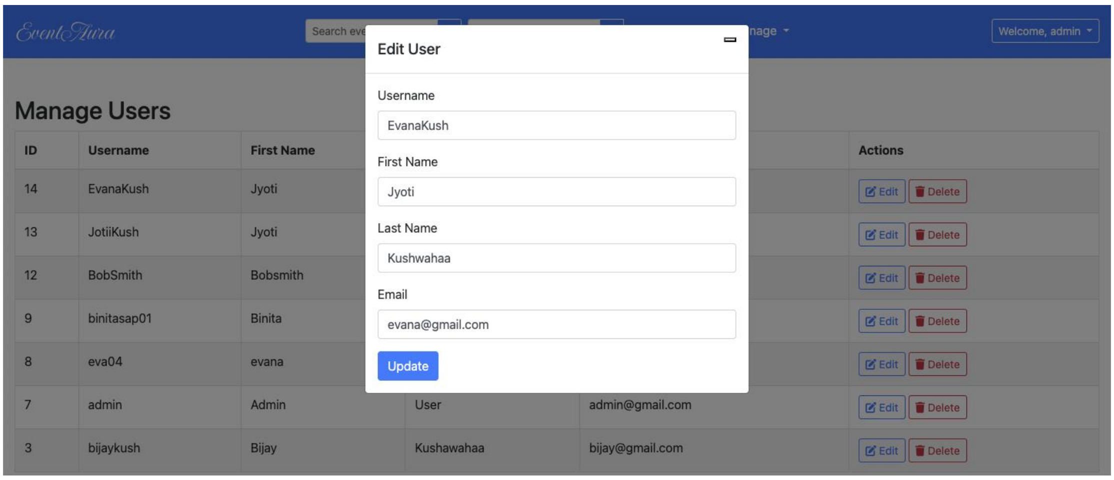
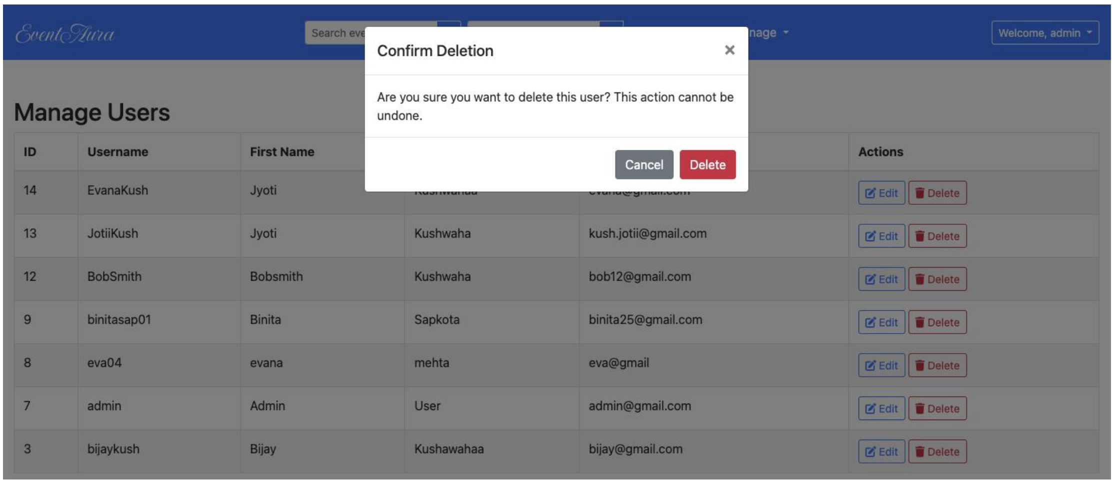


Figure 32: Admin Home Page







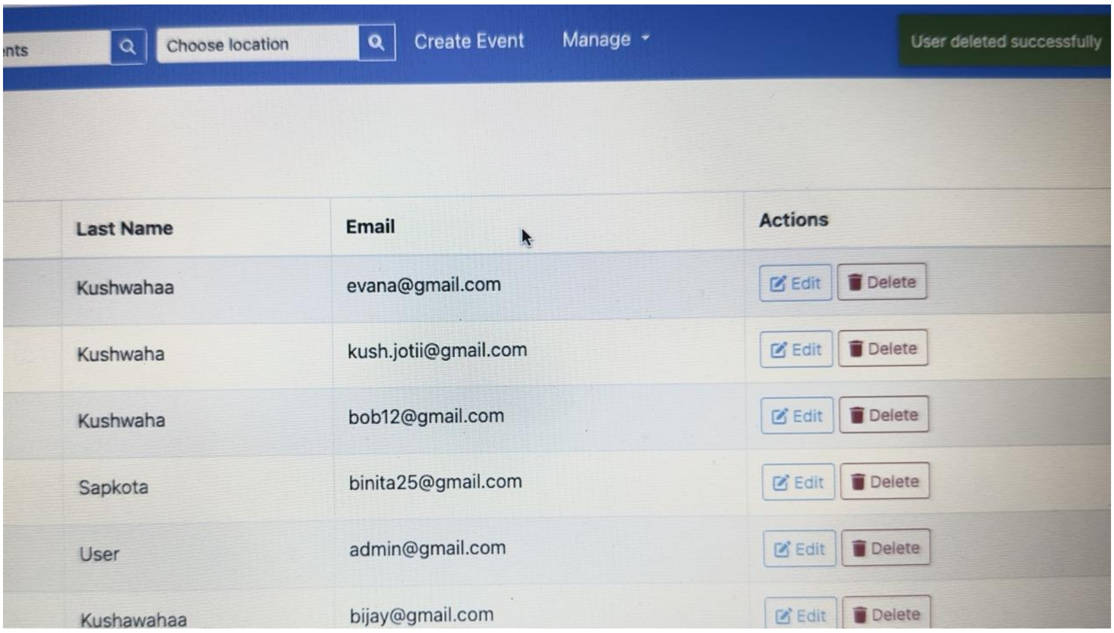
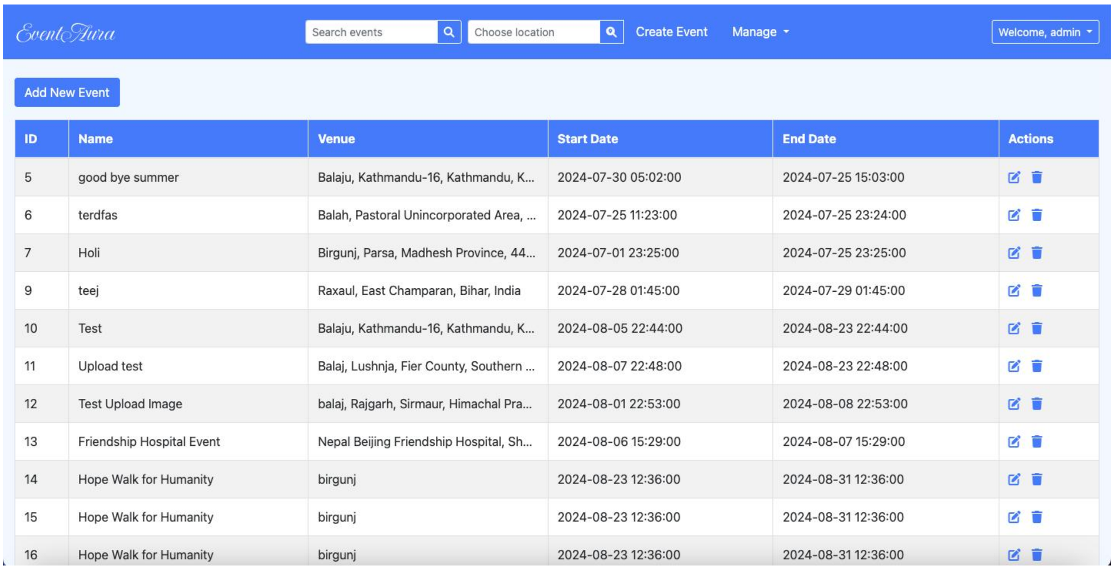
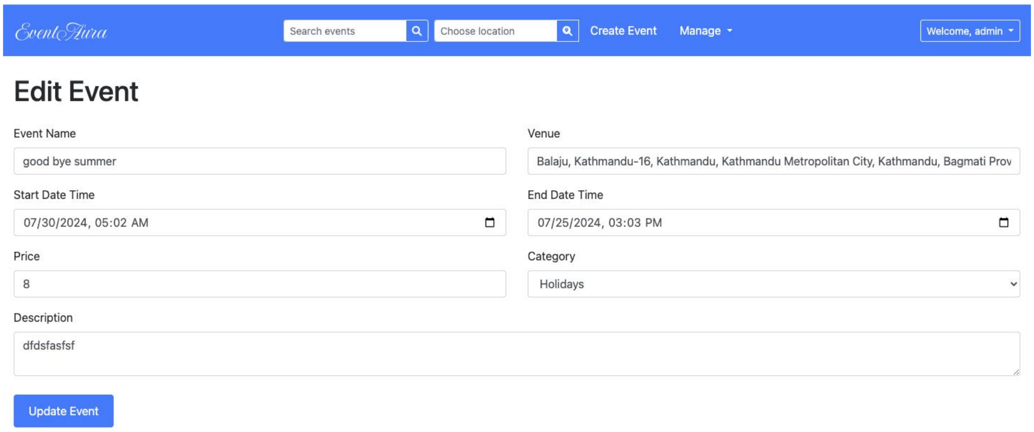


Figure 33: Manage User Page





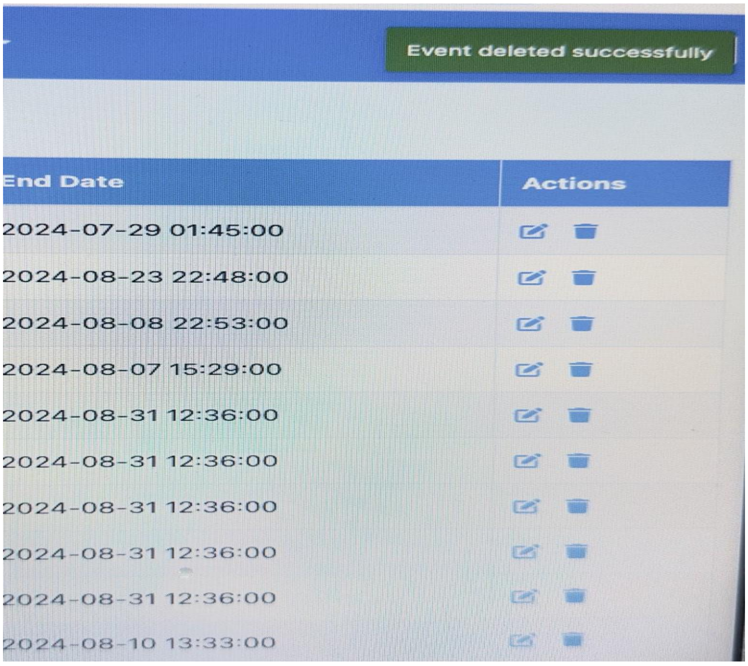


Figure 34: Manage Event Page

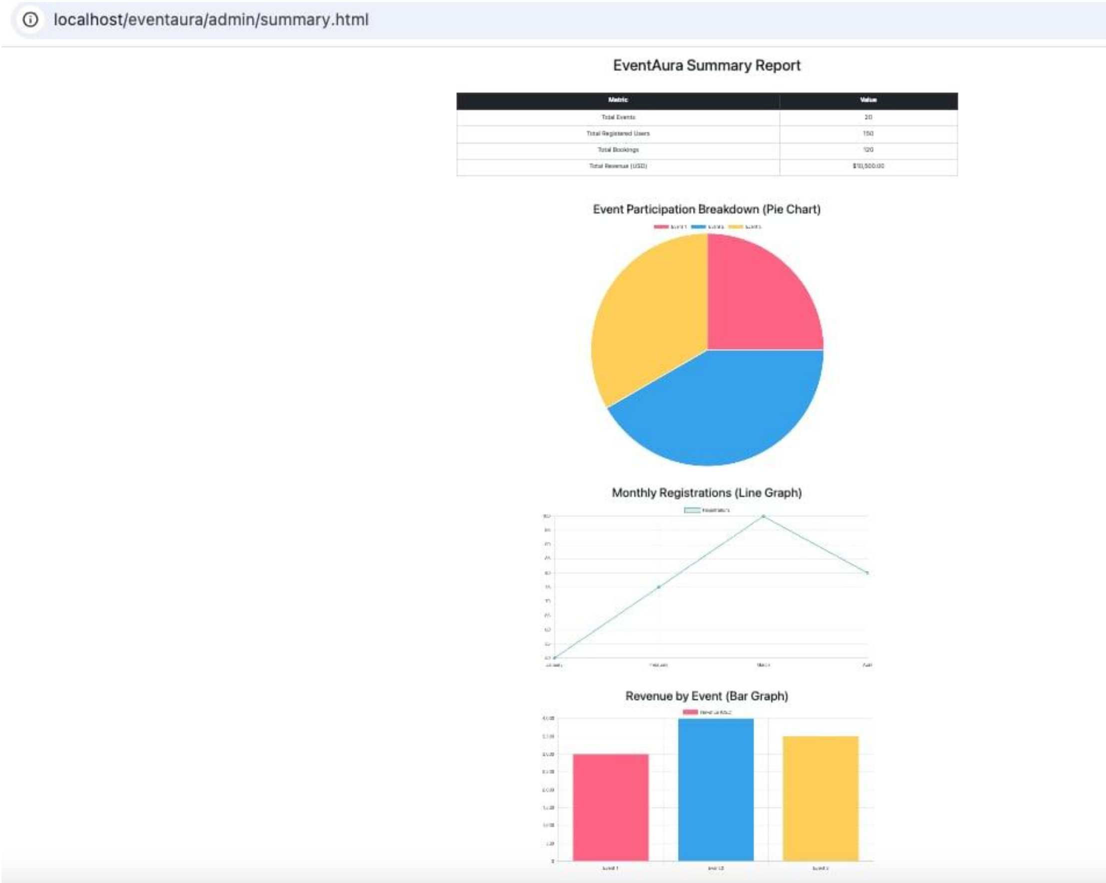


Figure 35: Summary Report

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